TaylorFlow Evaluation

Manual

19 October 2026 · version f63c080

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1 Installation

The program, its settings and its console messages use German names (for example `config.py`, `auswertung.py`, `MESSDATEIEN`). This manual explains every name in English; the names themselves are written exactly as they must be typed.

1.1 Installing Python

Python 3.10 or newer is required (tested with Python 3.12) together with these packages. They are installed with a single command in section 1.4:

Package	Used for
numpy, pandas, scipy	calculation, tables, curve fitting
matplotlib	all plots
thermo	property data of water (density, viscosity, surface tension, vapour pressure)
pyyaml	exclusion rules and own models (`.yaml` files)
pytest	checking the installation

The file selection window uses `tkinter`. It is part of the Python installation from python.org and does not need to be installed separately.

New Python installation (Windows)

1. Download the installer of the current version 3.12 or 3.13 from https://www.python.org/downloads/ (*Windows installer (64-bit)*).
2. Start the installer. In the first window **tick “Add python.exe to PATH” at the bottom**, then choose **Install Now**.
3. The default settings already include *tcl/tk and IDLE*, which the file selection window needs. Do not untick it under **Customize installation**.
4. Finish the installation.

Check

Open the command prompt (Start menu → “cmd”) and enter:

```
python --version
```

The expected answer is e.g. `Python 3.12.7`. If the Microsoft Store opens or Python is “not found”, the PATH option is missing: start the installer again, choose **Modify** → **Next** and tick *Add Python to environment variables*.

1.2 Downloading the program

The program is on GitHub: https://github.com/Oxymoron9946/TaylorFlow_Auswertung (branch `main`). The repository is public; no GitHub account is needed to download it.

Option A: as a ZIP file, no further software

1. Open the page in a browser.
2. Click **Code → Download ZIP** at the top right.
3. Unpack the ZIP file, e.g. to `C:\Measurements\TaylorFlow_Auswertung`. This folder is called the **project folder** below; it contains among others `config.py`, `auswertung.py` and `abbildung.py`.

For a newer version, download the ZIP file again and replace the folder. Back up your own changes to `config.py`, `config/ausschluesse.yaml` and `config/eigene_modelle.yaml` first.

Option B: with Git (convenient for later updates)

1. Install *Git for Windows* from https://git-scm.com/download/win (default settings are sufficient).
2. Open the command prompt, change to the desired parent folder and fetch the repository:

   ```
   cd C:\Measurements
   git clone https://github.com/Oxymoron9946/TaylorFlow_Auswertung.git
   ```

3. Fetch a newer version later (in the project folder):

   ```
   cd C:\Measurements\TaylorFlow_Auswertung
   git pull
   ```

   If `config.py` was changed locally, `git pull` only reports a conflict if the same file was also changed on GitHub.

1.3 Providing the measurement data

The evaluation reads the result files of the **video analysis** (manual of the video analysis, chapter 4.2):

File of the video analysis	Used here for
`output/Output_MassTransfer/Auswertung_Masse_<…>.csv`	mass transfer, `auswertung.py`
`output/Output_PressureCalibration/Auswertung_Druck_<…>.csv`	pressure calibration, `kalibrierung.py`

The files do not have to be copied: at start-up a file selection window opens in which they are chosen directly in the output folder of the video analysis. To make the window open there, enter the folder in `config.py`:

```
DIALOG_ORDNER = Path(r"C:\Measurements\Videoauswertung\output\Output_
 ↪  MassTransfer")
```

The measurement data of the existing test series are already in the project folder under `data/raw/` and can be used to try everything out. File structure: chapter 2.

1.4 Setting up

A separate Python environment (*virtual environment*, folder `.venv`) is created in the project folder. It contains exactly the required packages and does not interfere with any other Python installation.

1. Open the command prompt and change to the project folder:

   ```
   cd C:\Measurements\TaylorFlow_Auswertung
   ```

2. Create the environment (once) and activate it:

   ```
   python -m venv .venv
   .venv\Scripts\activate
   ```

   The prompt then starts with `(.venv)`. The activation only applies to this window; every new window needs `.venv\Scripts\activate` again.

3. Install the packages (once, takes one to two minutes):

   ```
   pip install -r requirements.txt
   ```

4. Check the installation:

   ```
   python -m pytest
   ```

   Expected: a line such as `40 passed`, no `failed`. The test evaluates the supplied measurement data completely and compares every number with the reference results in `tests/referenz/`. If it passes, the installation calculates correctly.

On Linux and macOS the activation is `source .venv/bin/activate`.

1.5 First start

With the environment activated, in the project folder:

```
python auswertung.py
```

The file selection window opens in `data/raw/`. Select one or more files `Auswertung_Masse_….csv` (several with the Ctrl key held down) and click **Open**. After about 20 seconds per file, tables and plots are in `ergebnisse/` (chapter 3).

1.6 Working in VS Code or Spyder

The scripts can also be started from a development environment. In both cases **the project folder must be the working directory**, otherwise `config.py` is not found.

VS Code

1. Install VS Code from https://code.visualstudio.com and add the extension *Python* (by Microsoft).
2. **File** → **Open Folder** → project folder.
3. At the bottom right or with Ctrl+Shift+P → *Python: Select Interpreter*, choose the environment `.venv`.
4. Open `auswertung.py` and click ▷ **Run Python File** at the top right. The output appears in the terminal below.

Spyder

1. Under **Tools → Preferences → Python interpreter**, enter the interpreter `…\TaylorFlow_Auswertung\.venv\Scripts\python.exe`. In the environment, also run `pip install spyder-kernels`.
2. **Run → Configuration per file** → *Working directory: the directory of the file being executed*.

1.7 Common errors

Message	Cause	Remedy
`'python' is not recognized …` or the Microsoft Store opens	Python is not on the PATH	installer → Modify → *Add Python to environment variables* (1.1)
`ModuleNotFoundError: No module named 'thermo'` (or `numpy`, `yaml`)	environment not activated or packages not installed	`.venv\Scripts\activate`, then `pip install -r requirements.txt`
`ModuleNotFoundError: No module named 'config'`	script not started from the project folder	change to the project folder (1.4) or set the working directory (1.6)
`ModuleNotFoundError: No module named 'tkinter'`	Python installed without *tcl/tk*	installer → Modify → tick *tcl/tk and IDLE*, or use a fixed file list instead of the selection window (chapter 3.5)
`Keine Messdatei ausgewählt, Auswertung beendet.`	selection window cancelled	start again and select a file
`Dateiname '…' passt nicht zum Schema`	file name does not follow the naming scheme	chapter 2.2
`In '…' fehlen die Spalten […]`	file does not come from the video analysis or is incomplete	chapter 2.3
tests fail with very small deviations	other package versions return property data that differ in the last digit	use the versions from `requirements.txt`; tested with numpy 2.5, pandas 3.0, scipy 1.18, thermo 0.6

2 Measurement data

2.1 Origin

The input data are the result files of the video analysis (see its manual, chapter 4). There is one file per channel and type of measurement; each row is one position bin of one camera position at one operating point:

```
Videoauswertung\output\
  Output_MassTransfer\
    Auswertung_Masse_PVC_R_1000um_72cm_20molCO2_600um.csv     →  auswertung.py
  Output_PressureCalibration\
    Auswertung_Druck_PVC_R_1000um_72cm_0molCO2_600um.csv      →  kalibrierung.py
```

The video analysis already writes file names and columns in the required format. The files are used unchanged and selected in the file selection window at start-up (chapter 3).

2.2 File names

All information on the experimental set-up is taken **only from the file name**; columns of the same name inside the file (Material, Kanal_um, …) are not used.

```
Auswertung_<Messart>_<Material>_<Form>_<D>um_<L>cm_<X>mol<Gas>_<Nadel>um.csv
```

Field	Meaning	Example
Messart	`Masse` = mass transfer measurement, `Druck` = measurement without CO2 for the pressure calibration	`Masse`
Material	channel material, free text (letters and digits)	`PVC`, `Glas`, `PTFE`
Form	`R` or `Rund` = circular channel; `Q` or `Eckig` = square channel	`R`
D	channel diameter or side length [µm]	`1000`
L	reactor length [cm]; sets the length of the calculation grid	`72`
X	CO2 fraction in the inlet gas [mol-%]	`20`
Gas	gas name; Henry constants are available for CO2, O2, N2 and air	`CO2`
Nadel	inner diameter of the gas inlet needle [µm]	`600`

The file name defines the **channel name**. Results, plots and exclusion rules are stored under this name:

```
File:          Auswertung_Masse_PVC_R_1000um_72cm_20molCO2_600um.csv
Channel name:  PVC_Rund_1000um_72cm_20molCO2_600um
```

Two series with the same channel name overwrite each other. A repeated measurement on the same set-up therefore gets its own material code, e.g. `PVC2`.

The pressure calibration uses a shorter **calibration key** without length and CO2 fraction, e.g. `PVC_circular_1000um_600um`. A pressure measurement thus applies to all mass transfer measurements on the same channel with the same needle.

2.3 File content

Format: semicolon as separator, decimal point or decimal comma, UTF-8 encoding.

Required columns

Column	Unit	Meaning
`GasVelocity`	m/s	superficial gas velocity u_G (set point)
`LiquidVelocity`	m/s	superficial liquid velocity u_L
`CameraPosition`	–	number of the camera position (1 = at the inlet)
`BinCenter_mm`	mm	position of the bin in the channel, measured from the inlet
`mean_BubbleLength_mm`	mm	mean bubble length in the bin
`mean_UnitCellLength_mm`	mm	mean length of the unit cell (bubble + slug)
`mean_Speed_m_s`	m/s	mean bubble velocity
`Temperatur_C`	°C	temperature (averaged per operating point)
`Druck_Pa_Auslass`	Pa	absolute pressure at the outlet

Also used, if present

Column	Use
`std_BubbleLength_mm`	measurement uncertainty in the bubble length fit and in the camera filter
`GroupCount` or `numel_BubbleLength_mm`	number of data points in the bin; for the standard error and the camera filter

Without the number per bin, the camera filter (2.5) is skipped and the fit weights with the standard deviation instead of the standard error.

Several recordings or sensitivities. If a file contains several rows for the same camera and the same bin, e.g. from several recordings or a sensitivity sweep of the video analysis, the row with the **smallest bubble length** is used per camera and bin. For a traceable evaluation, a file should contain only one sensitivity per operating point.

2.4 Exclusion rules

Data points or whole operating points that are not to be evaluated are listed in `config/ausschluesse.yaml`, grouped by channel name. Each rule states its reason (`grund`). There are three kinds:

```yaml
PVC_Rund_1000um_72cm_20molCO2_800um:
  messpunkte_entfernen:          # remove rows of the file before the evaluation
    - GasVelocity: 0.134
      LiquidVelocity: 0.25
      grund: Absorption vor dem ersten Messpunkt abgeschlossen
    - GasVelocity: 0.07
      LiquidVelocity: 0.111
      CameraPosition: 3          # only camera 3 of this operating point
      grund: Kamera 3 unscharf
  betriebspunkte_ueberspringen:  # operating point does not appear in any result
↪   file
    - LiquidVelocity: 0.056      # all operating points with u_L = 0.056 m/s
      grund: Pumpe instabil
  kla_ungueltig:                 # evaluated, but kLa_log_mean stays empty
    Gas_0_07_Liq_0_111: "Absorption kaum auswertbar"
```

Kind	Fields	Effect
`messpunkte_entfernen` (remove data points)	`GasVelocity`, `LiquidVelocity`, `CameraPosition`, `BinCenter_mm` (tolerance 0.1 mm); all given fields must match	rows are removed before the evaluation
`betriebspunkte_ueberspringen` (skip operating points)	`GasVelocity` and/or `LiquidVelocity`; a missing field means “all values”	operating point is not evaluated
`kla_ungueltig` (kLa invalid)	name of the operating point as in column `Config`	kLa_log_mean empty, `manual_exclude = True` with reason; profiles and bubble length fit remain available, 3D surfaces and model comparisons omit the point

The supplied file contains the exclusions of the supplied series with their reasons. After a change, run `auswertung.py` again.

2.5 Automatic camera filter

Before the evaluation, unreliable camera positions are removed (only for `Masse` files). A bin is valid if all three conditions are met:

- relative standard error of the bubble length ≤ 5 %
- at least 50 data points in the bin
- coefficient of variation ≤ 50 %

A camera of an operating point is kept if at least 75 % of its bins and at least 3 bins are valid. Operating points with only one remaining camera are dropped. The console reports how many bins and cameras were rejected. The limits are in `config.PARAMETER` (`kamerafilter_*`, chapter 5).

2.6 Evaluating a new series

1. Run the batch for the measurement folder in the video analysis. The file `Auswertung_Masse_….csv` is written to `output/Output_MassTransfer/`.

2. Start `python auswertung.py` and select the file in the selection window.
3. Check the results (chapter 3.4): warning log, bubble length fits, overview of key figures.
4. Enter unusable data points or operating points in `config/ausschluesse.yaml` (2.4) and evaluate again.
5. **Circular channel:** without an entry in the calibration database, the pressure model uses the theoretical value c = 0.465; the console then reports “nicht kalibriert” (not calibrated). The channel can be calibrated with a pressure measurement (chapter 10). **Square channel:** no calibration needed.

Other dimensions (diameter, length, CO2 fraction): nothing else needs to be adjusted; everything comes from the file name.

Liquids other than water and **rectangular, non-square channels** are not supported (chapters 8 and 12).

3 First evaluation

This chapter walks through a complete evaluation once, from the start to checking the results.

3.1 Program parts at a glance

All settings are in **one** file, `config.py`. One of five scripts in the project folder is started:

Script	Purpose	Chapter
`auswertung.py`	evaluate measurement files, create tables and plots	3, 4
`abbildung.py`	create and adjust a single figure, e.g. for a thesis	4.11
`literaturvergleich.py`	compare measurements with literature models, fit an own model	9
`kalibrierung.py`	determine the Bretherton factor c from pressure measurements	10
`monolith.py`	evaluate the monolith and compare it with a channel	11

3.2 Starting

In the command prompt (environment activated, chapter 1.4):

```
cd C:\Measurements\TaylorFlow_Auswertung
.venv\Scripts\activate
python auswertung.py
```

or open `auswertung.py` in VS Code and click ▷ **Run Python File**.

3.3 Selecting the measurement files

The window **“Messdateien auswählen”** (select measurement files) opens. It starts in the folder entered as `DIALOG_ORDNER` in `config.py` and shows only files `Auswertung_Masse_*.csv`. The list at the bottom right can also show all CSV files.

- One file: click it and **Open**.
- Several files: click them with the **Ctrl** key held down, or select all with **Ctrl+A**.
- **Cancel** ends the program without evaluation.

The console then shows the selected files and the main intermediate results for each operating point (program messages are in German):

```
Messdateien:
  C:\…\data\raw\Auswertung_Masse_Glas_Q_1000um_72cm_20molCO2_600um.csv

=== Glas_Eckig_1000um_72cm_20molCO2_600um ===
  200 Messpunkte laut ausschluesse.yaml entfernt.
  Kamerafilter: 0 von 2892 Bins entfernt (0.0 %), 0 von 54 Kamerapositionen.

  Betriebspunkt Gas_0_07_Liq_0_02 (u_G = 0.07 m/s, u_L = 0.02 m/s)
    übersprungen (ausschluesse.yaml): Starke Abweichung des gemessenen Drucks.

  Betriebspunkt Gas_0_07_Liq_0_111 (u_G = 0.07 m/s, u_L = 0.111 m/s)
    Blasenfit: Exp+Expansion v2+Offsets (…), R² = 0.8338, chi²_red = 5.06
    Offset Kamera 2: -0.024 mm
    Offset Kamera 3: -0.019 mm
    Druck: Auslass 100373.3 Pa, Einlass 108958.7 Pa
    kLa_log_mean = 0.285 1/s (721 Punkte in der aktiven Zone)
  …
Plots für Glas_Eckig_1000um_72cm_20molCO2_600um
  gespeichert in C:\…\ergebnisse\Glas_Eckig_1000um_72cm_20molCO2_600um\plots

Fertig nach 18 s.
```

Per operating point: bubble fit with R^2 and χ^2_red, camera offsets, outlet and inlet pressure (`Druck: Auslass …, Einlass …`) and kLa_log_mean with the number of grid points in the active zone.

3.4 Finding and checking the results

```
ergebnisse\
  Modell_Gueltigkeitswarnungen.txt          warnings of this run (chapter 6.4)
  Glas_Eckig_1000um_72cm_20molCO2_600um\
    …_betriebspunkte.csv                    one row per operating point, incl.
     ↪  kLa_log_mean
    …_profile.csv                           profiles along the channel
    …_meta.json                             source file and time
    plots\                                  all plots (chapter 4)
```

The CSV files can be opened directly in Excel (semicolon, decimal comma). Chapter 6 explains all columns.

For a new series, check in this order:

1. Read the **warning log** `Modell_Gueltigkeitswarnungen.txt` (meaning of the messages: chapter 6.4).
2. Look at the **bubble length fit** of each operating point (files `blasenlaenge_*.png` in `plots`, chapter 4.2): does the dark blue curve follow the data points of all cameras? Does the green curve decrease monotonically?
3. Check the **overview of key figures** `plots\heatmaps.png` for outliers.
4. Enter unusable points in `config/ausschluesse.yaml` (chapter 2.4) and evaluate again.

3.5 Common variants

Only redraw the plots, e.g. after a change in `PLOTS` or `STIL`: set in `config.py`

```
NEU_RECHNEN = False
```

and start `auswertung.py` again. Select the same files in the selection window; nothing is recalculated and drawing only takes seconds.

Without the selection window, e.g. for repeated runs with the same files:

```
MESSDATEIEN = [
    Path(r"C:\Measurements\Videoauswertung\output\Output_MassTransfer\Auswertung_
↪   Masse_PVC_R_1000um_72cm_20molCO2_600um.csv"),
    Path(r"C:\Measurements\Videoauswertung\output\Output_MassTransfer\Auswertung_
↪   Masse_Glas_Q_1000um_72cm_20molCO2_600um.csv"),
]
```

or `MESSDATEIEN = "alle"` for all files `Auswertung_Masse_*.csv` in `ROHDATEN_ORDNER` (default `data/raw/`).

Plots only for selected operating points:

```
BETRIEBSPUNKTE = ["Gas_0_134_Liq_0_056", "Gas_0_198_Liq_0_111"]
```

The names are in the column `Config` of the operating point table and in the console output.

4 Selecting and adjusting plots

4.1 Principle

After the calculation, `auswertung.py` creates all plots that are switched on and saves them in `ergebnisse/<channel>/plots/`. What appears and how it looks is set in two places in `config.py`:

Section	Applies to	Examples
`PLOTS`	one block per plot	on/off, axis ranges, plot elements, colours, colour map, heat map fields
`STIL` (style)	all plots together	language, file format, resolution, font size, default sizes, titles on/off

Each block in `PLOTS` lists **all** possible entries with their default value. To adjust a plot, only the value is changed:

```
PLOTS = dict(
    blasenlaengen_fit=dict(
        an=True, titel="auto", groesse_cm=None,
        x_bereich_cm=None,              # reactor position z [cm]
        y_bereich_mm=None,              # bubble length [mm]
        ...
```

After a change, a run with `NEU_RECHNEN = False` is sufficient (chapter 3.5); the plots are then redrawn within seconds.

Common to all plots:

Entry	Meaning	Example
`an` (on)	create the plot	`an=False`
`titel` (title)	`"auto"` = channel name (or channel and operating point), own text, or `""` for no title	`titel="PVC, 1 mm"`
`groesse_cm` (size)	(width, height) in cm; `None` = default size from `STIL`	`groesse_cm=(9, 7)`
axis ranges (`…_bereich…`)	`(min, max)` in the unit of the axis; `None` = automatic	`x_bereich_cm=(0, 40)`
colours (`farbe`)	name (`"tab:red"`, `"black"`) or hex code (`"#3B8B3B"`)	`farbe="tab:red"`
colour maps (`farbskala`)	name of a matplotlib colour map	`farbskala="viridis"`

Common colour maps: `"plasma"`, `"viridis"`, `"cividis"`, `"magma"`, `"coolwarm"`, `"RdYlGn"`, `"Blues"`. The suffix `_r` reverses a map (`"viridis_r"`).

Typing errors are reported immediately, with all possible names:

```
KeyError: "Unbekannte Option ['farbskale'] für den Plot 'kla_flaeche'.
Möglich: ['achsen', 'an', 'blickwinkel', 'farbskala', 'groesse_cm', ...]"
```

Overview of the plots

Block in PLOTS	Plot	File	Section
`blasenlaengen_fit`	bubble length along the channel with fit	`blasenlaenge_<operating point>.png`	4.2
`stoffmenge`	absorbed amount of CO2 per bubble	`stoffmenge_<operating point>.png`	4.3
`druckprofil`	calculated pressure along the channel	`druck_<operating point>.png`	4.4
`kla_flaeche`	kLa_log_mean over u_G and u_L (3D)	`kla_flaeche.png`	4.5
`nspez_flaeche`	volume-specific molar flux (3D)	`nspez_flaeche.png`	4.6
`modell_flaechen`	prediction of literature or own models (3D)	`modell_<model>.png`	4.7
`spalten_flaechen`	any column of the result table (3D), off by default	`flaeche_<column>.png`	4.8
`heatmaps`	overview of key figures as colour maps	`heatmaps.png`	4.9

The first three plots are created **per operating point**; which operating points is set by `BETRIEBSPUNKTE` (chapter 3.5). The others are created **per channel**.

4.2 Bubble length fit (`blasenlaengen_fit`)

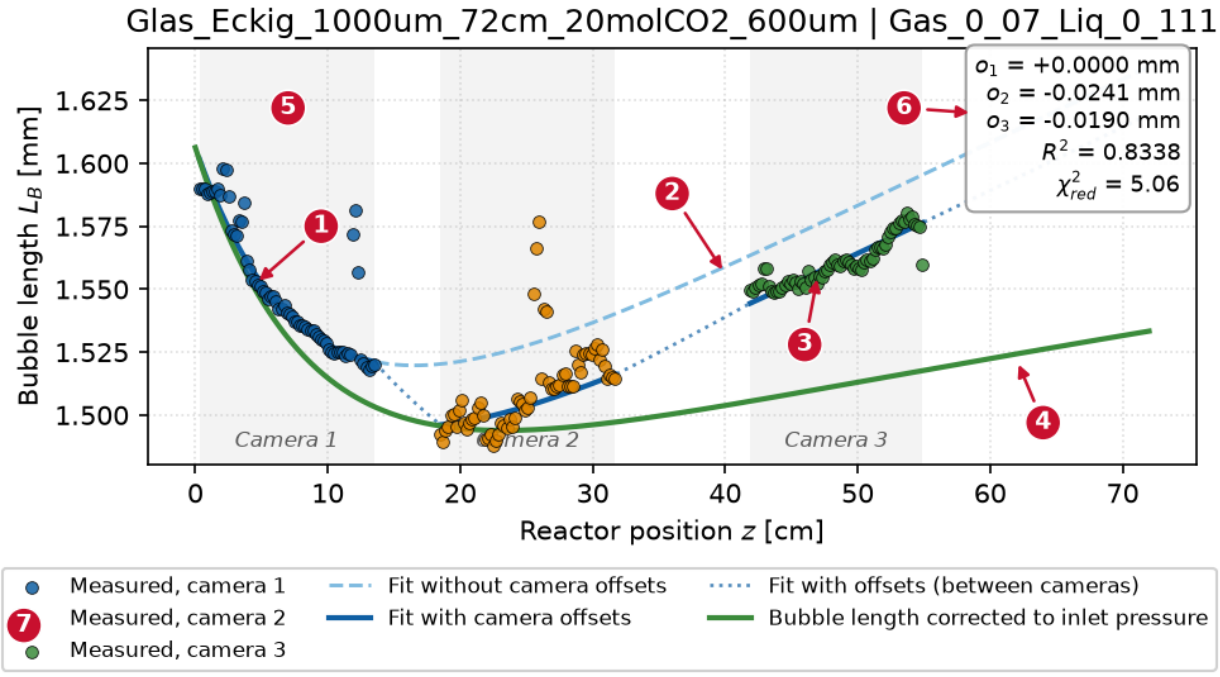


Figure 1: Bubble length fit with its plot elements, glass square 1000 µm, u_G = 0.07 m/s, u_L = 0.111 m/s

No.	Plot element	Meaning	Entry
1	dots, coloured per camera	measured bubble lengths per bin	`messwerte`
2	light blue, dashed	fit **without** camera offsets; this curve is used for the calculation	`fit_ohne_offsets`
3	dark blue, dotted between the cameras	the same fit plus the offset of each camera; must follow the dots	`fit_mit_offsets`
4	green	bubble length converted to the inlet pressure; contains only the absorption	`druckkorrigiert`
5	grey bands	range of the data points per camera	`kamerafenster`
6	box	camera offsets o_j, R^2 and χ^2_red of the fit	`kennwertkasten`
7	legend	explanation of the lines	`legende`

What to look for:

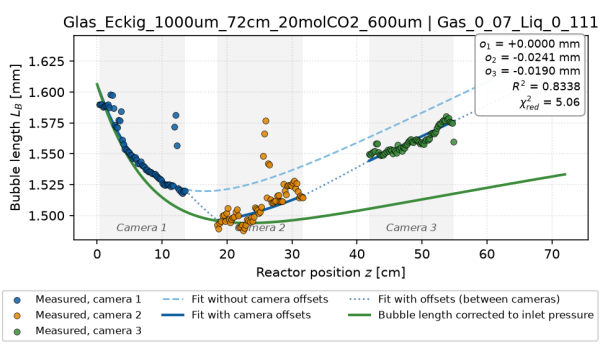
- The dark blue curve follows the dots of all cameras.
- The green curve decreases and levels off. A rise indicates a pressure profile that is calculated too low.
- The offsets are a few hundredths of a millimetre. If an offset is of the size of the absorption, the warning log reports it.
- χ^2_red above about 3 means that the model does not fully explain the scatter. Single outliers (in the figure at z ≈ 12 cm and 26 cm) are damped by the robust fit.

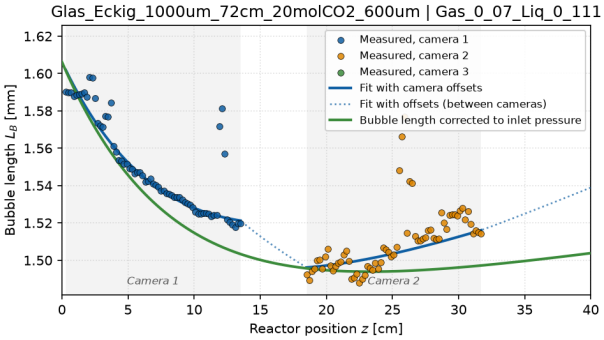
Adjusting

Wish	Entry in block `blasenlaengen_fit`
show only the front part of the channel	`x_bereich_cm=(0, 40)`
fixed y axis for several operating points	`y_bereich_mm=(1.4, 1.7)`
without the dashed fit	`fit_ohne_offsets=False`
data and fit only, without pressure correction	`druckkorrigiert=False`
without grey camera bands	`kamerafenster=False`
without the box of key figures	`kennwertkasten=False`
legend inside the plot instead of below	`legende="innen"` (free spot) or e.g. `legende="upper right"`
no legend	`legende="aus"`

Possible legend positions: `"upper left"`, `"upper center"`, `"upper right"`, `"center left"`, `"center"`, `"center right"`, `"lower left"`, `"lower center"`, `"lower right"`.

Example: left the default, right with `x_bereich_cm=(0, 40), fit_ohne_offsets=False, kennwertkasten=False, legende="upper right"`.



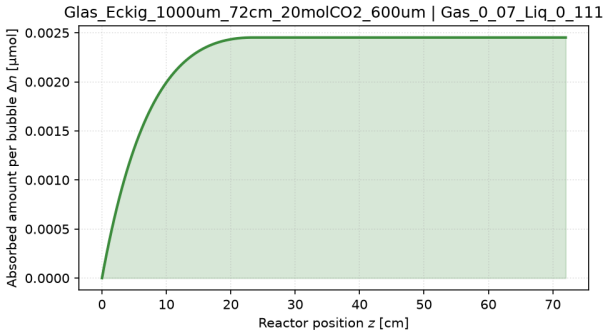


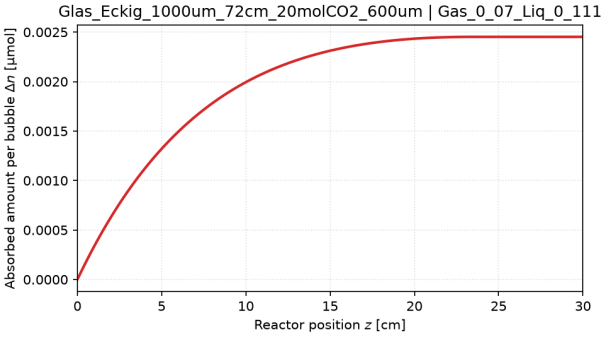
4.3 Absorbed amount (`stoffmenge`)

Cumulative absorbed amount of CO2 per bubble in µmol over the channel position. The curve levels off as soon as the liquid is saturated or the CO2 is used up.

Wish	Entry in block `stoffmenge`
axis ranges	`x_bereich_cm=(0, 30)`, `y_bereich_umol=(0, 0.5)`
no filled area below the curve	`flaeche=False`
other line colour	`farbe="tab:red"`

Left the default, right with `flaeche=False, farbe="tab:red", x_bereich_cm=(0, 30)`:



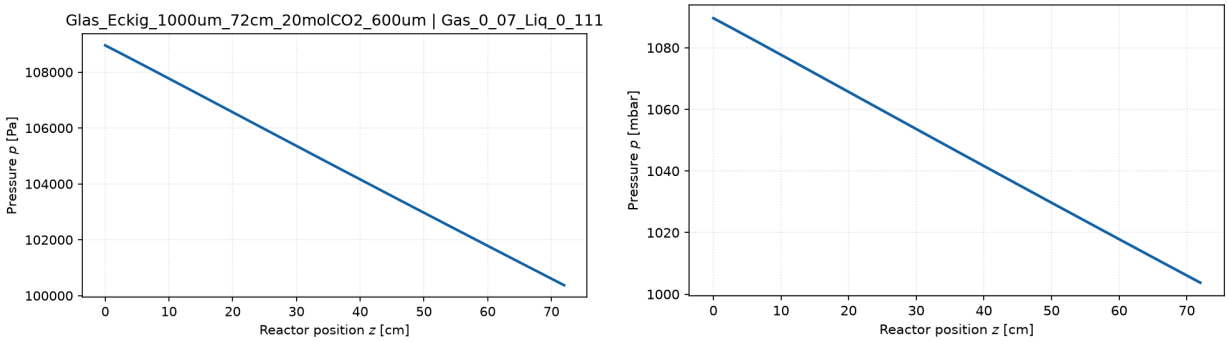


4.4 Pressure profile (`druckprofil`)

Calculated absolute pressure, integrated backwards from the measured outlet pressure. The slope is the pressure gradient of the Bretherton or Kurimoto model (chapter 7.4).

Wish	Entry in block `druckprofil`
pressure in mbar, hPa, kPa or bar	`einheit="mbar"`
axis ranges (y in the chosen unit)	`x_bereich_cm=(0, 72)`, `y_bereich=(1000, 1100)`
other line colour	`farbe="black"`

Left the default, right with `einheit="mbar", titel=""`:



4.5 kLa surface (`kla_flaeche`)

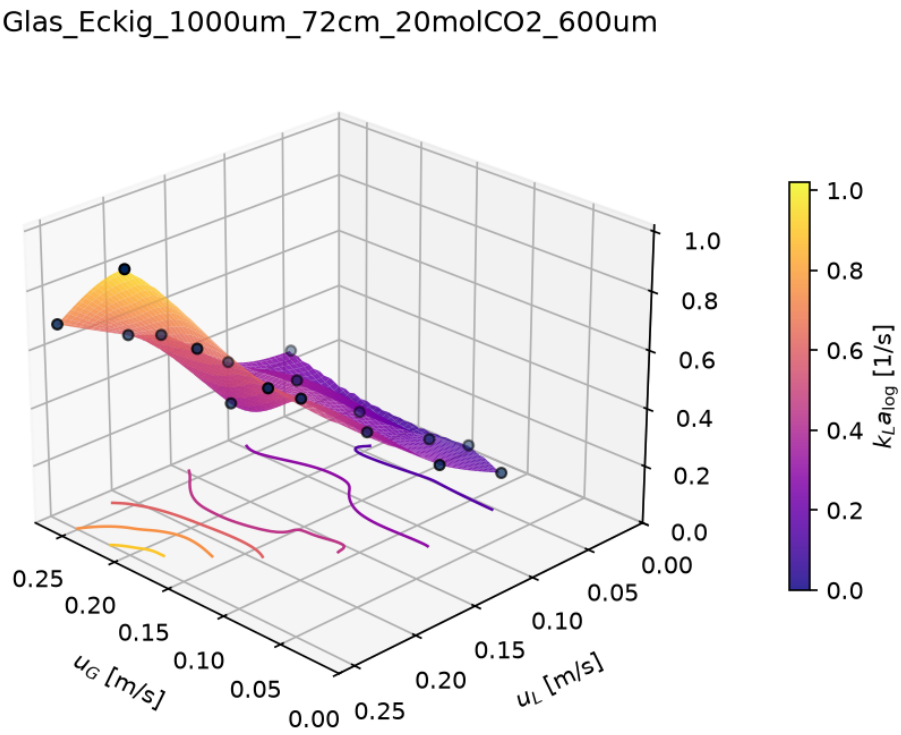


Figure 2: kLa_log_mean over u_G and u_L, glass square 1000 µm

- **Surface:** kLa_log_mean over u_G and u_L, interpolated cubically between the data points; outside the data points it stays empty.
- **Dots:** the measured values. With few, irregularly spaced points the interpolation can over- or undershoot; the dots are what counts.
- **Lines at the bottom:** contour lines.
- Operating points without a valid kLa value (`kla_ungueltig`) are missing.

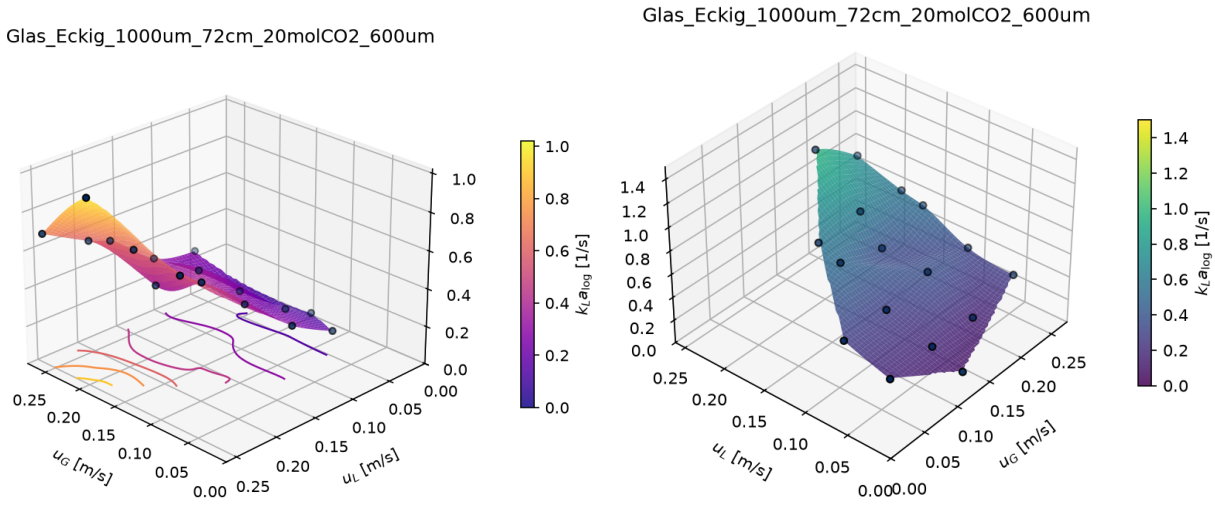
Adjusting (applies equally to all 3D surfaces, sections 4.6 to 4.8)

Wish	Entry
other colour map	`farbskala="viridis"`

Wish	Entry
fixed axis ranges, e.g. for comparing several channels	`achsen=dict(x=(0, 0.3), y=(0, 0.3), z=(0, 1.5))`; axes not given remain automatic
other viewing direction	`blickwinkel=(35, 220)`: elevation and rotation in degrees
without data points	`messpunkte=False`
without contour lines	`hoehenlinien=False`
opaque or more transparent surface	`transparenz=1.0` or `0.5`

Viewing direction: elevation 0° is the side view, 90° the top view; the rotation turns the plot about the vertical axis (default `(25, 135)`).

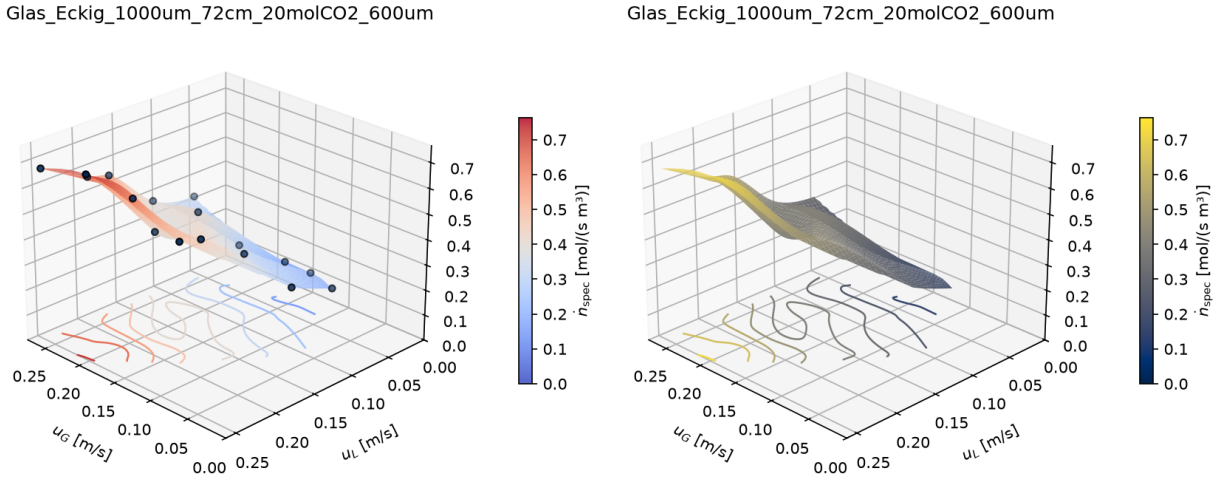
Left the default, right with `farbskala="viridis", blickwinkel=(35, 220), achsen=dict(z=(0, 1.5)), hoehenlinien=False`:



4.6 Volume-specific molar flux (`nspez_flaeche`)

ṅ_spec [mol/(s·m³)]: absorbed molar flux of CO2 related to the whole channel volume (definition in chapter 7.7). Relating it to the whole channel makes the value comparable with the monolith; saturated sections therefore count as well. Options as in 4.5.

Left the default, right with `farbskala="cividis", messpunkte=False, transparenz=1.0`:



4.7 Model surfaces (`modell_flaechen`)

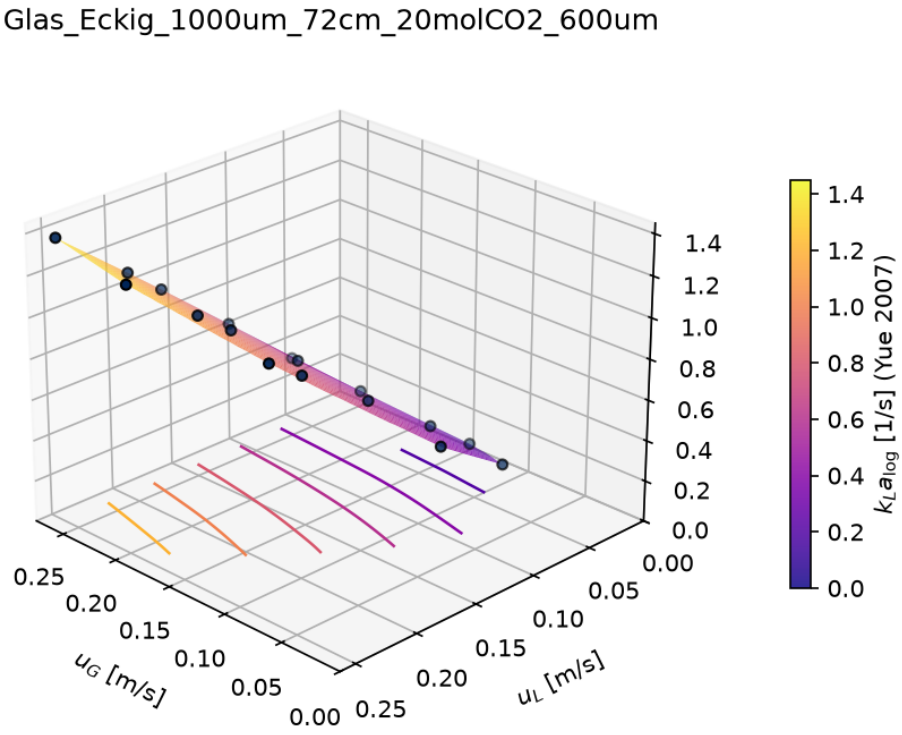


Figure 3: Prediction of Yue 2007 for the glass channel

Prediction of a literature or own model for the same operating points, one figure per model. The models are listed in `modelle`:

```
modell_flaechen=dict(an=True, modelle=["yue_2007", "vandu"], ...)
```

Possible names: `van_baten_voll`, `van_baten_kappe`, `van_baten_film`, `vandu`, `shao`, `liu_wang`, `heiszwolf`, `yao`, `yue_2007`, `yue_2009` and all own models from `config/eigene_modelle.yaml` (chapter 9). Models that do not apply to the channel shape (Yue square only, Liu-Wang circular only) are skipped with a note. Other options as in 4.5.

4.8 Any column as a surface (`spalten_flaechen`)

Every numeric column of the operating point table (chapter 6.2) can be drawn as a 3D surface over u_G and u_L. The block is switched off by default. Each entry in `spalten` creates one figure `flaeche_<column>.png`:

```
spalten_flaechen=dict(
    an=True, ...,
    spalten=[
        dict(spalte="SlugLength_mean_m", beschriftung="Slug length $L_S$ [mm]",
         ↪  faktor=1000),
        dict(spalte="slip_meas_over_theo", beschriftung="Slip [-]"),
    ],
)
```

Entry per column	Meaning
`spalte` (column)	column name as in the table
`beschriftung` (label)	text at the colour bar; without it the column name. Formulas in `$…$`
`faktor` (factor)	conversion before drawing, e.g. `1000` for m → mm

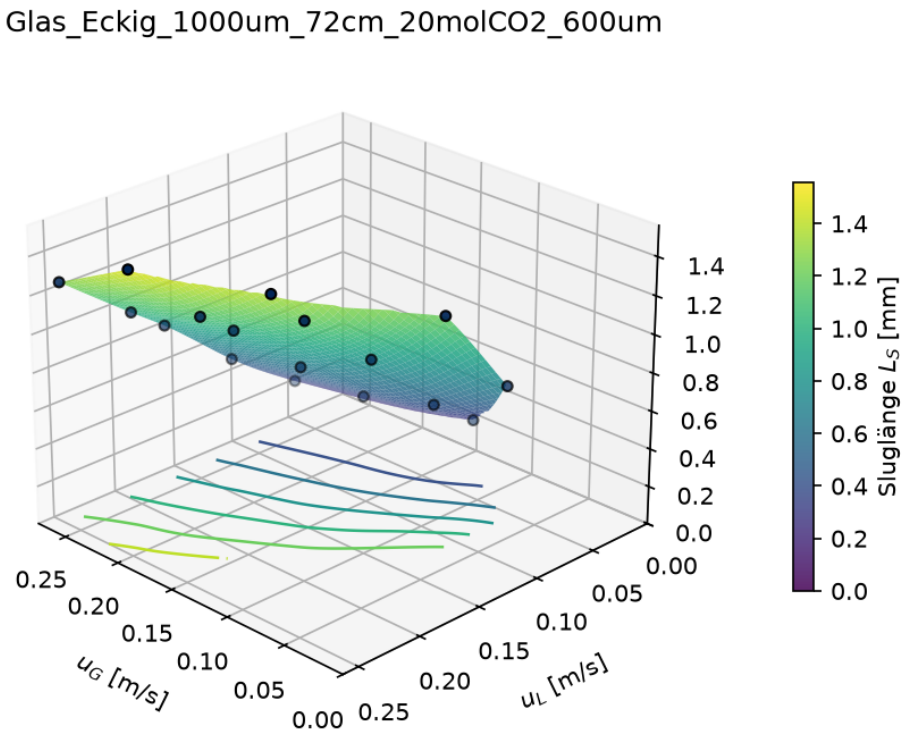


Figure 4: Slug length as a surface

4.9 Overview of key figures (`heatmaps`)

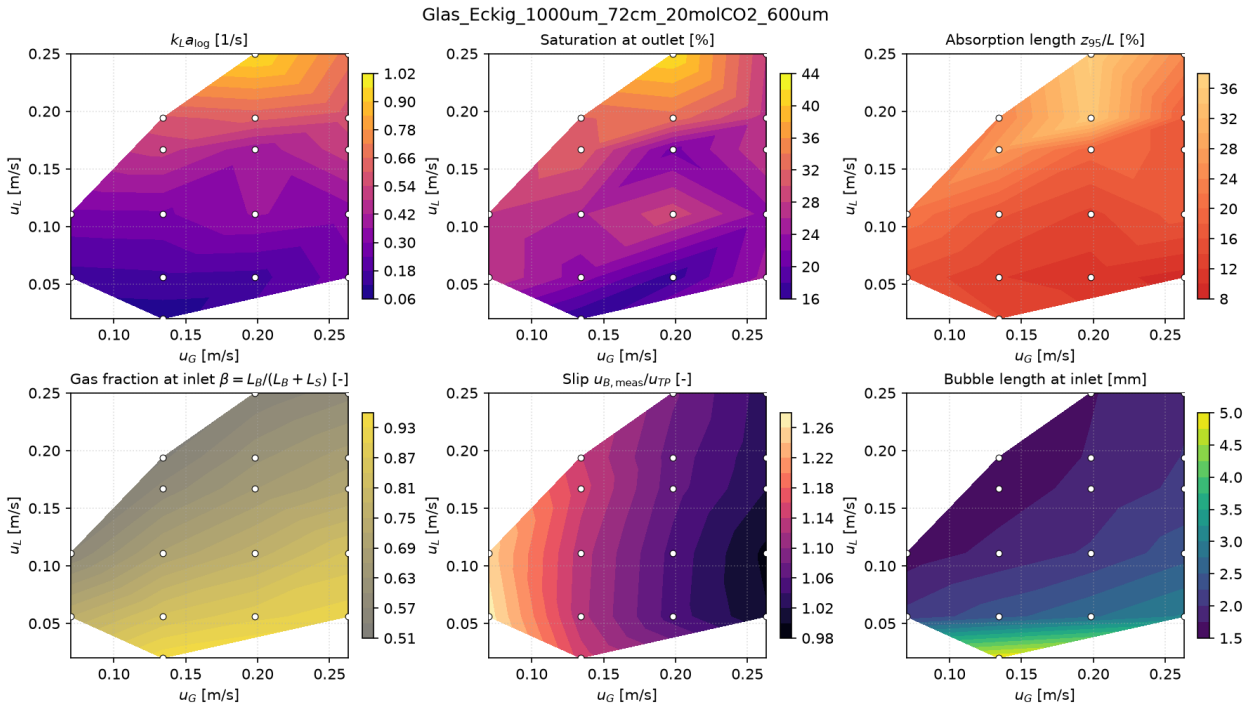


Figure 5: Overview of key figures, glass square 1000 µm

Several quantities as colour maps over u_G × u_L; the white dots are the operating points. The overview is suited to checking a new series quickly for outliers. Default fields in reading order:

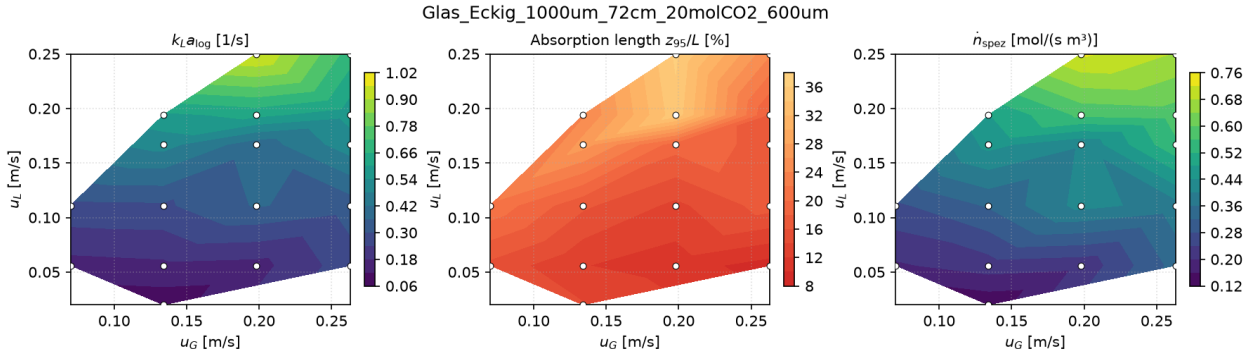
Name in `felder`	Quantity
`"kla"`	kLa_log_mean
`"saettigung_aus"`	saturation C_L/C* at the channel end
`"z95"`	absorption length: position at which 95 % of the total absorbed amount of CO2 has been taken up, relative to the channel length
`"beta_ein"`	gas fraction of the unit cell at the inlet, β = L_B/(L_B + L_S)
`"schlupf"`	measured bubble velocity / (u_G + u_L)
`"L_b_ein"`	fitted bubble length at the inlet

Adjusting

Wish	Entry in block `heatmaps`
other selection or order	`felder=["kla", "z95", "schlupf"]`; up to three fields per row
a column of the operating point table as a field	`felder=["kla", "n_spez_mol_s_m3"]`
heading of a field	`beschriftungen={"n_spez_mol_s_m3": "ṅ_spec [mol/(s m³)]"}`
colour map of a field	`farbskalen=dict(kla="viridis")`

Wish	Entry in block `heatmaps`

Example with three fields: `felder=["kla", "z95", "n_spez_mol_s_m3"]`, own heading for `n_spez_mol_s_m3` and `farbskalen=dict(kla="viridis")`:

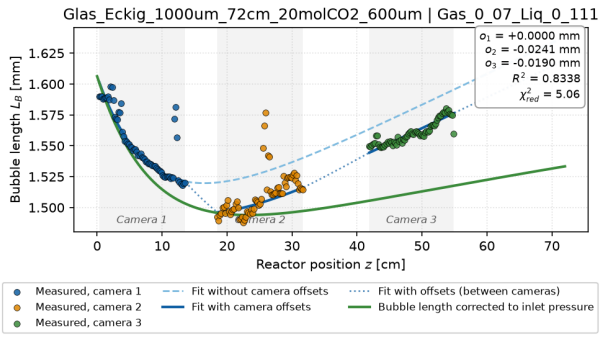


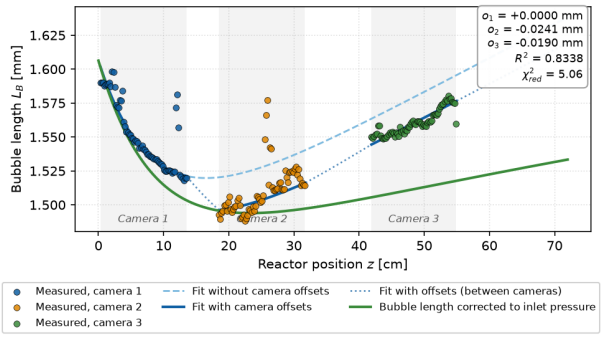
4.10 Appearance of all plots (`STIL`)

Entry	Default	Meaning
`sprache` (language)	`"de"`	`"de"` (decimal comma) or `"en"` (decimal point, English labels)
`formate` (formats)	`("png",)`	one or more file formats, e.g. `("png", "pdf")`; PDF and SVG scale without loss
`dpi`	300	resolution of the PNG files
`schriftgroesse` (font size)	10	base font size in pt
`groesse_profil_cm`	(16, 9)	default size of the plots per operating point
`groesse_3d_cm`	(14, 11)	default size of the 3D surfaces
`groesse_heatmaps_cm`	(30, 17)	default size of the key figure overview (for 2 × 3 fields)
`groesse_paritaet_cm`	(12, 11)	parity plots (chapters 9 and 11)
`titel_anzeigen` (show titles)	`True`	`False` switches off all titles, e.g. for figures in a thesis
`blickwinkel_3d`	(25, 135)	default viewing direction of the 3D surfaces

`STIL` also applies to `literaturvergleich.py`, `monolith.py` and `kalibrierung.py`. A size in the block of a plot (`groesse_cm`) takes precedence over the default size. All figures in this English manual were created with `sprache="en"`.

Left with title (default), right with `titel_anzeigen=False`:





4.11 Creating a single figure (`abbildung.py`)

A figure for a thesis or a presentation often needs exactly one plot in a particular form, without changing `config.py` permanently. This is what `abbildung.py` is for. At the top of the file there are only a few settings:

```
KANAL = "Glas_Eckig_1000um_72cm_20molCO2_600um"   # folder name in ergebnisse/
PLOT = "blasenlaengen_fit"                        # name of a block in config.PLOTS
BETRIEBSPUNKT = "Gas_0_134_Liq_0_056"             # only for plots per operating
 ↪  point

ANPASSUNGEN = dict(                               # changes compared with
 ↪  config.PLOTS[PLOT]
    x_bereich_cm=(0, 40),
    kennwertkasten=False,
    legende="upper right",
)
STIL_ANPASSUNGEN = dict(                          # changes compared with config.STIL
    sprache="en",
    formate=("png", "pdf"),
    titel_anzeigen=False,
)
ZIEL_ORDNER = config.ERGEBNIS_ORDNER / "abbildungen"   # target folder
DATEINAME = None        # file name; None = name as in auswertung.py
VORSCHAU = True         # also show the figure in a window
```

Then run `python abbildung.py`. The figure is saved and, with `VORSCHAU = True`, shown in a window. In this window it can be zoomed and moved; the saved version is the one with the settings from the file.

- Nothing is recalculated. `auswertung.py` must already have been run for the channel.
- `ANPASSUNGEN` accepts all entries of the respective block in `config.PLOTS` (sections 4.2 to 4.9).
- For `modell_flaechen` and `spalten_flaechen`, one figure is created per model or column.

4.12 Recipes

Figure for a thesis (narrow, English, vector graphics). In `abbildung.py`:

```
PLOT = "kla_flaeche"
ANPASSUNGEN = dict(groesse_cm=(8, 7), farbskala="viridis")
STIL_ANPASSUNGEN = dict(sprache="en", formate=("pdf",), titel_anzeigen=False,
 ↪  schriftgroesse=8)
```

Comparing several channels on the same axes. Set fixed axes for the surface in `config.py`, then evaluate or redraw all channels:

```
kla_flaeche=dict(..., achsen=dict(x=(0, 0.3), y=(0, 0.3), z=(0, 1.5)), ...)
```

Only the overview plots, no plots per operating point. In `config.py`, set `an=False` in the blocks `blasenlaengen_fit`, `stoffmenge` and `druckprofil`.

Another quantity as a surface. In `spalten_flaechen`, set `an=True` and enter the column in `spalten` (4.8). Chapter 6.2 lists the available columns.

Key figure overview with an own selection. `felder` in block `heatmaps` (4.9).

4.13 Plots directly from Python

All plot functions can also be called directly. They return a matplotlib figure that can be edited further before saving:

```
import config
from taylorflow import ergebnisse, plots

ordner = config.ERGEBNIS_ORDNER
kanal = "Glas_Eckig_1000um_72cm_20molCO2_600um"
bp = ergebnisse.lade_betriebspunkte(ordner, kanal)
profile = ergebnisse.lade_profile(ordner, kanal)
stil = plots.PlotStil(**config.STIL).mit(sprache="en")

fig = plots.kla_flaeche(bp, stil, titel="", farbskala="viridis", achsen=dict(z=(0,
 ↪  1.5)))
fig.axes[0].set_zlabel("kLa [1/s]")              # any matplotlib change
plots.speichere(fig, "abbildungen", "kla_glas", stil)

profil = ergebnisse.profil_eines_betriebspunkts(profile, "Gas_0_134_Liq_0_056")
zeile = bp[bp["Config"] == "Gas_0_134_Liq_0_056"].iloc[0]
fig = plots.blasenlaengen_fit(profil, zeile, stil, x_bereich_cm=(0, 40),
 ↪  legende="innen")
plots.speichere(fig, "abbildungen", "blasen_glas", stil)
```

The parameters of each function correspond to the entries of the blocks in `config.PLOTS`; `help(plots.blasenlaengen_fit)` shows them.

5 Configuration

All settings are in `config.py` in the project folder. The file is divided into numbered sections. Plots and appearance (sections 4 and 5 of the file) are described in chapter 4; this chapter describes the remaining entries.

Values that affect the calculation result are in `PARAMETER`. With their default values, the reference results in `tests/referenz/` are obtained. The column **Effect** states how strongly a change affects kLa_log_mean. It is based on variation runs with the supplied series (5 channels, 55 operating points); "–" means not investigated separately.

5.1 Folders and files

Entry	Default	Meaning
`ROHDATEN_ORDNER` (raw data folder)	`data/raw`	supplied measurement files; source for `MESSDATEIEN = "alle"`
`DIALOG_ORDNER` (dialog folder)	same as `ROHDATEN_ORDNER`	folder in which the selection window starts, e.g. the output folder of the video analysis
`ERGEBNIS_ORDNER` (results folder)	`ergebnisse`	outputs; one subfolder per channel
`AUSSCHLUSS_DATEI` (exclusion file)	`config/ausschluesse.yaml`	exclusion rules (chapter 2.4)
`KALIBRIER_DATENBANK` (calibration database)	`data/calibrated_c_database.json`	Bretherton factors of the circular channels (chapter 10)
`MODELL_DATEI` (model file)	`config/eigene_modelle.yaml`	saved own models (chapter 9)

Paths outside the project folder are given as `Path(r"…")`, e.g.

```
DIALOG_ORDNER = Path(r"D:\Videoauswertung\output\Output_MassTransfer")
```

5.2 Scope of the evaluation

Entry	Default	Meaning
`MESSDATEIEN` (measurement files)	`"dialog"`	`"dialog"` = selection window; `"alle"` = all `Auswertung_Masse_*.csv` in `ROHDATEN_ORDNER`; or a list of file paths
`NEU_RECHNEN` (recalculate)	`True`	`False`: only redraw the plots from existing results
`BETRIEBSPUNKTE` (operating points)	`None`	operating points for the plots per operating point; `None` = all

5.3 Evaluation parameters (PARAMETER)

5.3.1 Camera filter

Entry	Default	Unit	Meaning
kamerafilter_rel_se_max_prozent	5.0	%	max. relative standard error of the bubble length per bin
kamerafilter_n_min	50	–	min. number of data points per bin
kamerafilter_cv_max_prozent	50.0	%	max. coefficient of variation per bin
kamerafilter_anteil_min	0.75	–	min. fraction of valid bins for a camera to be kept
kamerafilter_bins_min	3	–	min. number of valid bins per camera

5.3.2 Bubble length fit

Entry	Default	Unit	Meaning	Effect
fit_serien_offsets	True	–	fit camera offsets o_j (camera 1 is the reference)	–
fit_loss	"soft_l1"	–	robust loss function; "linear" = classical least squares	–
sigma_sys_m	2e-6	m	systematic measurement error, added in quadrature to the standard error	–
sigma_sem_floor_m	1e-6	m	lower limit of the standard error per bin	–
fit_lambda0_m	0.05	m	start value of the absorption length λ	none (0.01–0.2 m)
offset_bound_anteil	0.2	–	limit |o_j| ≤ fraction · mean bubble length	none (limit never reached)

The camera resolves about 55 µm/px. The two error limits of 1 and 2 µm are far below this; they only prevent bins with very many bubbles from dominating the fit.

5.3.3 Integration

Entry	Default	Unit	Meaning	Effect
dz_m	1e-3	m	step size for pressure profile and mass balance	small (0.25–2 mm: a few per cent)
u_fit_cap_faktor	1.5	–	fitted velocities above factor · U_ref are discarded and interpolated; U_ref = max(median of the measurement, u_G + u_L)	none (never active)

Entry	Default	Unit	Meaning	Effect
`saettigung_clip`	0.999	–	C_L is limited to this fraction of C*	none (0.99–0.999)

5.3.4 Measurement uncertainties

Entry	Default	Meaning
`u_rel_h_cp`	0.03	relative uncertainty of the Henry constant
`u_rel_p_co2`	0.02	relative uncertainty of the CO2 partial pressure

Both values determine where the active zone ends (chapter 7.6). The other three entries of this section (`k_opt_mm_pro_px`, `u_rel_dn_dt`, `pixel_unsicherheit_px`) belong to the error propagation in `taylorflow/unsicherheit.py`. This function is available for own uncertainty analyses and is not called by the evaluation.

5.3.5 kLa evaluation

Entry	Default	Meaning	Effect
`max_urel_delta_c`	0.5	a grid point belongs to the active zone if ΔC > 0 and the uncertainty of C* is at most this fraction of ΔC	none (0.3–0.7)
`logmean_min_aktive_punkte`	10	fewer active grid points: kLa_log_mean = 0	–

5.3.6 Pressure drop and property data

Entry	Default	Meaning
`c_bretherton_theorie`	0.465	Bretherton factor for circular channels without a usable calibration
`fluessigkeit` (liquid)	`"Water"`	liquid name for the library `thermo`

The literature models use fixed property data for water/CO2 at about 25 °C (chapter 9.2). They are defined in `taylorflow/parameter.py` (`lit_*`) and can also be set in `PARAMETER` if required.

An unknown name in `PARAMETER` produces a message listing all permitted names.

5.4 Further sections

Section	Script	Chapter
`VERGLEICH` (comparison)	`literaturvergleich.py`	9

Section	Script	Chapter
KALIBRIERUNG (calibration)	kalibrierung.py	10
MONOLITH	monolith.py	11

6 Results

6.1 Course of the evaluation

For each selected measurement file, `auswertung.py` performs these steps (code: `taylorflow/pipeline.py`, equations: chapter 7):

```
measurement file ──► reading, exclusions, camera filter            (einlesen.py)
   │
   └─► per operating point (u_G, u_L):
         1  data points: one value per camera and bin
         2  property data at T and p at the outlet (thermo, Henry, vapour pressure)
         3  fits over z: bubble length (with camera offsets), slug length, velocity
         4  pressure profile: backward integration from the outlet (Bretherton or
          ↪  Kurimoto)
         5  mass balance: bubble volume → absorbed amount → C_L, C*, saturation
         6  active zone and kLa_log_mean
         7  bubble length converted to inlet pressure
         8  literature models
   │
   └─► saving: profiles, operating point table, metadata           (ergebnisse.py)
Then: plots according to config.PLOTS                          (taylorflow/plots/)
```

6.2 Result files

For each channel, `ergebnisse/<channel>/` is created with:

File	Content
`<channel>_betriebspunkte.csv`	one row per operating point: key figures, fit quality, literature models
`<channel>_profile.csv`	one row per grid point (1 mm) and operating point
`<channel>_meta.json`	program version, time, source file
`plots/`	plots according to `config.PLOTS` (chapter 4)

Format: semicolon, decimal comma, UTF-8; the files can be opened directly in Excel. In Python they are read with `taylorflow.ergebnisse.lade_betriebspunkte(folder, channel)` or `lade_profile(folder, channel)`; these functions read numbers without rounding errors. An existing file of the same name is overwritten.

Operating point table (`…_betriebspunkte.csv`)

Column	Unit	Meaning
`Config`	–	name of the operating point, e.g. `Gas_0_134_Liq_0_056`
`GasVelocity_m_s`, `LiquidVelocity_m_s`	m/s	superficial velocities u_G, u_L

Column	Unit	Meaning
kLa_log_mean_1_s	1/s	**main result:** log-mean volumetric mass transfer coefficient over the active zone (chapter 7.6); 0 if the active zone is too short, empty for kla_ungueltig
BubbleLength_mean_m, BubbleLength_start_m	m	fitted bubble length, mean over the channel and at the inlet
BubbleLength_meas_mean_m	m	mean of the measured bubble lengths
SlugLength_mean_m	m	fitted slug length, mean over the channel
Velocity_mean_m_s	m/s	fitted bubble velocity, mean; input of the literature models
Velocity_meas_mean_m_s	m/s	mean of the measured velocities
slip_meas_over_theo	–	median measured velocity / (u_G + u_L); usually 1.05–1.30
n_spez_mol_s_m3	mol/(s·m³)	volume-specific molar flux of CO2 over the whole channel (chapter 7.7)
fit_model	–	bubble model used; normally Exp+Expansion v2+Offsets …
fit_r2, fit_chi2_red	–	quality of the bubble fit; χ^2_red ≈ 1 ideal, > 3 conspicuous, > 10 critical
fit_corr_max	–	largest correlation of two fit parameters; > 0.95 = poorly determined
fit_n_bound_hits, fit_bound_hits	–	parameters at a fit limit (number, description)
fit_n_series, fit_n_offsets	–	number of cameras and of fitted offsets
fit_offset_o2_mm, fit_offset_o3_mm	mm	camera offsets relative to camera 1
kLa_lit_*_1_s	1/s	predictions of the literature models (chapter 9)
d_h_m, Geometrie	m, –	channel diameter or side length; rund (circular) or eckig (square)
D_CO2_m2_s	m²/s	diffusion coefficient of CO2 in water at T
T_sys_K, P_aus_Pa	K, Pa	temperature and outlet pressure (mean of the operating point)
manual_exclude, manual_exclude_reason	–	True and reason for an entry under kla_ungueltig

Profile table (…_profile.csv)

Column	Unit	Meaning
Config_Name	–	operating point
Position_z_m	m	position in the channel (grid with dz_m)
L_bubble_fit_m	m	fitted bubble length (without camera offsets)
L_bubble_corrected_m	m	bubble length converted to the inlet pressure; shows only the absorption

Column	Unit	Meaning
`L_slug_fit_m`, `U_fit_m_s`	m, m/s	fitted slug length and bubble velocity
`Pressure_Pa`	Pa	calculated absolute pressure
`Diffused_Gas_mol`	mol	amount of CO2 absorbed per bubble up to z (monotonic)
`C_L_bulk_mol_m3`	mol/m³	CO2 concentration in the liquid
`C_star_lokal_mol_m3`	mol/m³	equilibrium concentration C* = H · p_CO2
`Saturation_percent`	%	saturation C_L / C*
`C_star_lokal_raw_mol_m3`, `Saturation_percent_raw`	mol/m³, %	the same quantities before the monotonicity and saturation limits
`aktive_zone`	–	1 = grid point in the active zone (chapter 7.6), 0 = not
`fit_offset_o1_m` … `o3_m`	m	camera offsets (constant per operating point)
`cam1_z_min_m` … `cam3_z_max_m`	m	range of the data points per camera
`z_meas_m`, `L_b_meas_m`	m	data points (only in the first rows, empty otherwise)

6.3 Console output

For each operating point the console reports:

- the bubble model with R^2 and χ^2_red
- the camera offsets
- for circular channels, the Bretherton factor c used
- inlet and outlet pressure
- kLa_log_mean and the number of grid points in the active zone

6.4 Warning log

`ergebnisse/Modell_Gueltigkeitswarnungen.txt` is rewritten in every run. It collects, separated by channel and operating point:

Message	Meaning	Action
`Liu (2005): … außerhalb`	operating point outside the range of the slip correlation	usually none, note only
`Bretherton (1961): Ca_L … > 0.005`	capillary number above the validity range of the Bretherton pressure model	treat the pressure profile with caution
`Kurimoto: …`	Ca, We, Re or L_B/d outside the Kurimoto range	treat the pressure profile with caution
`Filmdicke Modell: …`	film thickness outside the model range, approximation used	check the operating point
`Blasen-Fit: chi2_red = … -> ERHEBLICH` or `KRITISCH`	the fit describes the measurement poorly	look at the bubble length fit (chapter 4.2)

Message	Meaning	Action
`Blasen-Fit: Parameter hoch korreliert`	A and λ (or B) cannot be determined independently	check the kLa of the point critically
`Serien-Offset … nahe/am Bound`	cameras strongly offset against each other	check the camera calibration
`Serien-Offset … > 50% des Absorptionssignals`	the offset may take up part of the absorption	check the plot, exclude the camera if necessary
`Slip U_meas/U_TP > 1.5`	measured velocity implausibly high	check the set points of the flow controllers
`kla_ungueltig aktiv: …`	entry from `ausschluesse.yaml` applied	–
`kla_ungueltig: Betriebspunkt '…' nicht gefunden`	name in `ausschluesse.yaml` matches no operating point	check the spelling (column `Config`)

7 Methodology

This chapter describes the chain of models as implemented in the code.

Symbols: z position from the inlet, L_B bubble length, L_S slug length, U bubble velocity, d channel diameter or side length, δ film thickness, T temperature, p pressure, C_L concentration in the liquid, C* equilibrium concentration.

7.1 Assumptions

- **Isothermal:** T and p at the outlet are averaged per operating point. Property data (ρ, μ, σ, H, D, p_v) at these values apply to the whole channel.
- **Gas phase:** ideal gas. The gas is immediately saturated with water vapour; the dry gas pressure is $p_{tr} = p - p_v$.
- **Absorption:** only CO2 is absorbed; the inert part (N2) stays in the bubble.
- **Liquid volume:** the liquid volume of a unit cell is determined at the inlet and remains constant.
- **Mixing:** the slug is ideally mixed (one concentration C_L per position).

7.2 Bubble length fit

The measured bubble lengths of all cameras are fitted with the model

$$L_B(z) = L_0 - A\left(1 - e^{-z/\lambda}\right) + B\,z \;+\; o_j$$

- A: shortening due to absorption
- λ: absorption length
- B: expansion due to the pressure drop
- o_j: constant offset of camera j relative to camera 1 (o_1 = 0), which absorbs calibration differences between the camera positions

The calculation continues with the curve **without** offsets.

Weighting:

$$\sigma = \sqrt{\sigma_{SEM}^2 + \sigma_{sys}^2}, \qquad \sigma_{SEM} = s/\sqrt{n} + 1\,\mu\text{m}, \quad \sigma_{sys} = 2\,\mu\text{m}.$$

Procedure:

1. Weighted fit.
2. Robust fit (soft-L1) with a scale from the MAD of the residuals of the first pass.

Parameter limits:

- L_0 ∈ [0.1; 5] · max(L)
- A ≤ 2 · range(L)

- λ between the smallest spacing of data points and 10 L_reactor
- 0 ≤ B ≤ 2 max(L)/L_reactor
- |o_j| ≤ 0.2 · mean bubble length

Fallback models: if the fit fails, a model without offsets follows, then $ae^{-bz} + c + dz$, finally a straight line. The name of the model used is in the column `fit_model`.

Slug length $L_S = L_{UC} - L_B$ and velocity U are fitted linearly over z. Fitted velocities above $1.5 \cdot U_{ref}$ are discarded and interpolated; $U_{ref} = \max(\text{measured median}, u_G + u_L)$.

7.3 Film thickness and bubble geometry

Capillary number $Ca = \mu U/\sigma$.

Channel	Validity	Film thickness
circular	Ca ≤ 0.005	Bretherton: $\delta = \frac{d}{2}\, 1.34\, Ca^{2/3}$
circular	0.005 < Ca ≤ 1.5	Aussillous/Quéré: $\delta = \frac{d}{2}\, \frac{1.34\, Ca^{2/3}}{1 + 3.35\, Ca^{2/3}}$
square	–	Kreutzer/Thulasidas: $\delta/d = 0.357 - 0.25\, e^{-2.25\, Ca^{0.445}}$

Bubble: diameter or side length $d_B = d - 2\delta$. It consists of a cylinder (circular) or cuboid (square) of length $L_B - d_B$ and two caps with a combined volume $V_{caps} = \frac{\pi}{6} d_B^3$:

$$V_B = V_{caps} + A_B \max(0,\, L_B - d_B), \qquad A_B = \tfrac{\pi}{4} d_B^2 \text{ (circular)}, \quad A_B = d_B^2 \text{ (square)}.$$

Liquid per unit cell (at the inlet), with the channel cross-section A_K:

$$V_L = L_S A_K + L_B (A_K - A_B)\ (+ V_{caps}/2 \text{ in the circular channel}).$$

7.4 Pressure profile

Backward integration from the measured outlet pressure with step dz = 1 mm: $p(z_i) = p(z_{i+1}) + (\mathrm{d}p/\mathrm{d}z)_i\, \Delta z$. In both models the fitted bubble velocity $U(z)$ is used as the velocity u_{TP}.

Circular channel (Bretherton 1961, after Fries 2008):

$$f_{app} = \frac{16}{Re}\left[1 + c\, \frac{d}{L_S}\, Ca^{-1/3}\right], \qquad \Delta p_{UC} = 2 f_{app}\, \rho_L u_{TP}^2 \frac{L_S}{d}, \qquad \frac{\mathrm{d}p}{\mathrm{d}z} = \frac{\Delta p_{UC}}{L_S + L_B}.$$

c comes from the calibration database (chapter 10):

- linear interpolation between the calibrated operating points
- outside them, the stored fallback model
- without a usable entry, the theoretical value c = 0.465

Square channel (Kurimoto et al. 2020, eq. 27/28):

$$\Delta p_B = \frac{\sigma}{d}\,\frac{7.106\,Ca^{2/3}}{N}, \qquad \frac{\mathrm{d}p}{\mathrm{d}z} = \frac{L_S}{L_{UC}}\,\frac{\rho_L u^2}{2d}\,\frac{56.9}{Re}\left[1 + \frac{d}{L_S}\,\frac{0.250\,Ca^{-1/3}}{N}\right],$$

with $N = 1 + 0.744\,Ca^{2/3} - 0.623\,We^{0.120}$. There is no free factor. Ca, Re and We are formed with the bubble velocity, as intended by Kurimoto.

7.5 Mass balance

Amount of gas in the bubble at position z and at the inlet:

$$n(z) = \frac{p_{tr}(z)\,V_B(z)}{R\,T}, \qquad n_0 = n(0).$$

Of this, $n_{CO_2,0} = y_{CO_2}\,n_0$ and $n_{inert} = (1 - y_{CO_2})\,n_0$. Absorbed up to z (limited to $[0,\ n_{CO_2,0}]$, then made monotonic):

$$\Delta n(z) = n_0 - n(z), \qquad C_L(z) = \Delta n(z)/V_L.$$

Equilibrium with the local CO2 fraction in the dry gas:

$$y(z) = \frac{n_{CO_2,0} - \Delta n}{n_{inert} + n_{CO_2,0} - \Delta n}, \qquad C^*(z) = H(T)\,p_{tr}(z)\,y(z),$$

$$H(T) = H_{298}\,\exp\!\left[\frac{\Delta_{sol}H}{R}\left(\frac{1}{T} - \frac{1}{298.15}\right)\right]$$

with $H_{298} = 3.36 \cdot 10^{-4}\,\mathrm{mol/(m^3\,Pa)}$ and $\Delta_{sol}H/R = 2400\,\mathrm{K}$ for CO2 (Sander 2015).

C_L is limited to $0.999\,C^*$. The saturation is C_L/C^*.

7.6 Active zone and kLa_log_mean

Active zone. A grid point belongs to it if the driving concentration difference $\Delta C = C^* - C_L$ is positive and larger than twice the uncertainty of C*:

$$\Delta C > 0 \quad \text{and} \quad \frac{u_{rel}(C^*)\,C^*}{\Delta C} \le 0.5, \qquad u_{rel}(C^*) = \sqrt{0.03^2 + 0.02^2} \approx 3.6\,\%.$$

When the liquid is nearly saturated, ΔC can no longer be determined reliably; there the active zone ends. The condition is met as long as the saturation stays below about 93 %; in the supplied series the active zone therefore usually covers almost the whole channel.

kLa_log_mean. For plug flow with constant kLa, $\mathrm{d}C_L/\mathrm{d}t = k_L a\,(C^* - C_L)$ holds. Integrated over the active zone, from the first up to and including the last active point:

$$k_L a_{log} = \frac{1}{t_{res}}\,\ln\frac{\Delta C_{in}}{\Delta C_{out}}, \qquad t_{res} = \int_{z_{in}}^{z_{out}} \frac{\mathrm{d}z}{U(z)}$$

(Kashid 2011).

- kLa_log_mean uses ΔC at the boundaries of the active zone and the residence time. It is therefore insensitive to noise in the values in between.
- It is 0 with fewer than 10 active grid points or if ΔC does not decrease in the active zone.
- It stays empty if the operating point is listed under `kla_ungueltig`.

7.7 Further key figures

Volume-specific molar flux over the whole channel (same reference as for the monolith):

$$\dot{n}_{spec} = \frac{\Delta n(z_{end})\, f_B}{A_K\, z_{end}}, \qquad f_B = \frac{U(0)}{L_B(0) + L_S(0)}$$

(f_B is the bubble frequency).

Pressure-corrected bubble length: the bubble volume is converted to the inlet pressure, $V_B\, p_{tr}(z)/p_{tr}(0)$, and back into a length. Without absorption the curve would be constant.

Absorption length z95 (key figure overview): position at which Δn reaches 95 % of its value at the channel end, relative to the channel length.

Slip: median of the measured bubble velocity / (u_G + u_L).

7.8 Measurement uncertainty

An error propagation for the mass transfer along z (uncertainties of molar flux, liquid volume and ΔC) is included as the function `relative_unsicherheit` in `taylorflow/unsicherheit.py`. The evaluation does not use it; it is available for own analyses.

7.9 References

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- Bretherton, F. P. (1961). The motion of long bubbles in tubes. *J. Fluid Mech.* 10, 166–188.
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- Versteeg, G. F., van Swaaij, W. P. M. (1988). Solubility and diffusivity of acid gases in aqueous alkanolamine solutions. *J. Chem. Eng. Data* 33, 29–34.

Sources of the literature models: chapter 9.

8 Limitations

8.1 Validity ranges of the models

Model	Range	Behaviour outside
Bretherton pressure drop (circular)	Ca ≤ 0.005	warning in the log, calculation continues
Kurimoto pressure drop (square)	0.00159 ≤ Ca ≤ 0.0989; 0.0817 ≤ We ≤ 25.4; 8.33 ≤ Re ≤ 791; L_B/d ≥ approx. 2.3	warning, calculation continues
film thickness circular	Ca ≤ 1.5	approximation δ = 0.05 d with warning
slip Liu (2005)	0.008–0.7 m/s (gas), 0.008–0.5 m/s (liquid), d 0.9–3 mm	warning (note only, not used in the calculation)
Henry constant	CO2, O2, N2, air in water	stop with message
CO2 diffusion	water	–
literature models	see chapter 9.2; the supplied channels lie outside the original range of almost all models	–

In the supplied glass channel, some operating points at low u_G fall slightly below the Kurimoto criterion L_B/d.

8.2 Known limitations

1. **Isothermal, property data at the outlet:**
   - T and p at the outlet apply to the whole channel.
   - For temperature changes of more than approx. 5 K along the channel, a section-wise evaluation would be needed.
2. **Constant liquid volume:**
   - V_L is determined at the inlet.
   - Changes of the slug length along z do not enter C_L.
3. **Caps and corners in the square channel:**
   - The liquid in the corners is added to the liquid volume (as half the cap volume) only in the circular channel.
   - The caps are approximated as half-ellipsoids in both geometries.
   - A cube instead of an ellipsoid approximation of the caps changes single kLa values in the glass channel by up to 26 % (median 1.8 %).
4. **Pressure model of the circular channels:**

- No published pressure model (Bretherton, Kreutzer 2005, Warnier 2010) reproduces all three supplied circular channels with less than 70 % mean deviation without channel-specific adjustment.
- Calibrated c values are therefore stored for PVC (chapter 10).
- Channel PVC 1500/1150 is marked as unusable (c between −7.6 and +16); c = 0.465 applies there.

5. **Square and circular channels compared:**
   - In the glass channel (square), the measured values follow the trend of the literature models, especially Yue 2007; the absolute values differ in part by a multiple.
   - The circular channels show hardly any trend. Likely causes are the pressure model and material effects.
6. **Active zone as one range:**
   - kLa_log_mean uses the first and the last active point.
   - If the active zone were interrupted, the sections in between would be counted in the residence time. This does not occur in the supplied series.
7. **Literature models at 25 °C:**
   - Ca, Re and Sc of the literature models are calculated with fixed property data for about 25 °C.
   - The rest of the evaluation uses the measured temperature.
   - At 20–25 °C the difference is a few per cent.
8. **Cache of the property data:**
   - Property data are cached with the temperature rounded to 0.1 K.
   - Operating points with almost the same temperature receive identical values. The result of an operating point therefore depends slightly on which points were calculated before in the same run: if an operating point is added or excluded, other kLa_log_mean values change by at most about 0.002 %.
   - The reference results apply to evaluating all supplied channels in alphabetical order.
9. **3D surfaces:**
   - The cubic interpolation can over- or undershoot with few, irregularly spaced points.
   - For quantitative statements, the data points count.
10. **Several recordings per bin:** if a file contains several rows for the same camera and bin, the smallest bubble length is used (chapter 2.3).

8.3 Sensitivity to the settings

All fixed thresholds were varied with the supplied series (5 channels, 55 operating points). Effect on kLa_log_mean:

Quantity	Variation	Effect
start value λ_0 of the fit	0.01–0.2 m	none (identical results)
step size dz	0.25–2 mm	a few per cent, below the measurement uncertainty
saturation limit	0.99–0.999	none

Quantity	Variation	Effect
limit of the active zone u_rel(ΔC)	0.3–0.7	none
cap on U_fit	–	never active
offset limit	–	never reached
fallback model of the calibration	linear instead of polynomial	≤ 3.6 % in one operating point
film thickness model	δ/d 0.03–0.10, Aussillous instead of Bretherton	none (approximation never active)

The results therefore depend practically not on the numerical settings. What matters are the measurement data, the pressure model and the exclusion rules.

8.4 Open points

- **Pressure model of circular channels:** a physically founded alternative to the calibration is missing (8.2 no. 4).
- **Calibration:** the fallback models are fitted with only 5–8 points (chapter 10.5).
- **Rectangular, non-square channels:** geometry, pressure and film models would have to be extended.
- **Other liquids:** Henry constant and diffusion coefficient would have to be added (chapter 12.3).

9 Literature comparison and own models

`python literaturvergleich.py` compares a measured quantity with literature models and fits an own model if desired.

9.1 Procedure

1. `auswertung.py` must have been run for the desired channels.
2. In `config.py`, section `VERGLEICH`, set:

Entry	Default	Meaning
`kanaele` (channels)	`None`	`None` = all channels with results; otherwise a list of channel names
`zielgroesse` (target quantity)	`"kLa_log_mean_1_s"`	any numeric column of the operating point table
`groesse_beschriftung` (label)	`r"$k_La_{\mathrm{log}}$ [1/s]"`	axis label of the target quantity
`modelle` (models)	all literature models	list of models for the table and the parity plots
`band_prozent` (band)	15	width of the band ± around the diagonal [%]
`logarithmisch` (logarithmic)	`True`	logarithmic axes
`eigener_fit` (own fit)	block	fit an own model (9.3); `None` = no fit

3. Run it. Outputs in `ergebnisse/vergleich/`:
   - `vergleich_<target>.csv`: statistics per model, sorted by MAPE
   - `paritaet_<model>.png`: parity plot per model, channels in colour
   - `eigenes_modell_<name>.png`: parity of the fitted own model

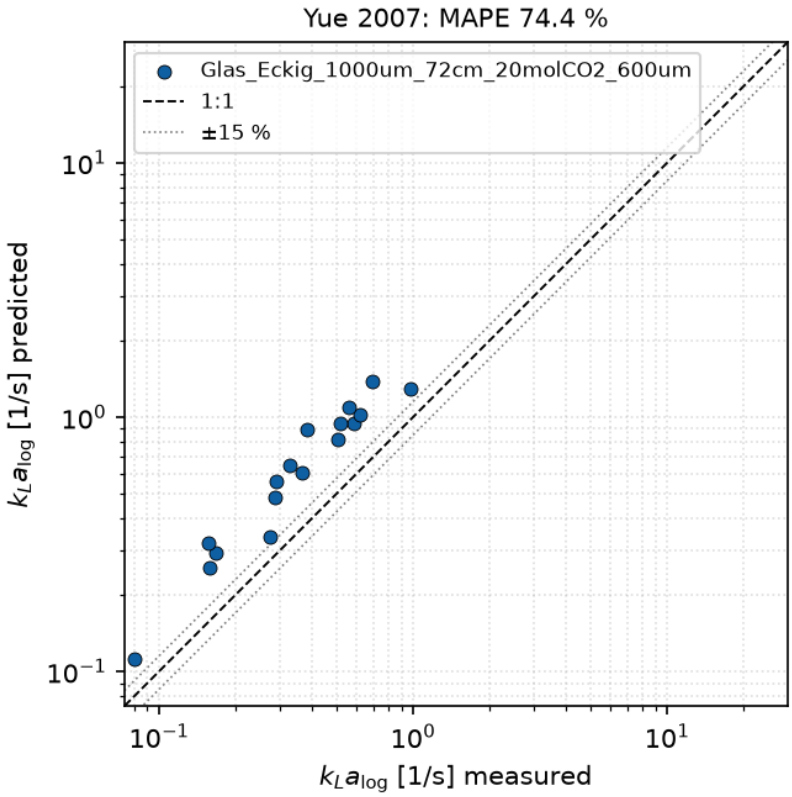


Figure 6: Parity plot Yue 2007 for the glass channel

Parity plot: measured against calculated values, channels with their own colour and marker. The dashed diagonal is perfect agreement; the dotted lines bound the deviation ±`band_prozent`. The title gives the MAPE. Channels to which the model does not apply (here the circular channels for Yue 2007) are missing. Size and language follow `STIL` (`groesse_paritaet_cm`, `sprache`, chapter 4.10).

Operating points with `manual_exclude = True` are not used.

Statistics in `vergleich_<target>.csv`:

Column	Meaning
`n`	number of points with a valid measured and model value
`MAPE_%`	mean absolute percentage error
`Bias_%`	median of the relative deviation; positive = model too high
`RMSE`	root mean square error (unit of the target quantity)
`im_Band_%`	fraction of points within ±`band_prozent`

9.2 Literature models

All models are in `taylorflow/literatur.py` with formula and original range. Their predictions are also included as columns `kLa_lit_*_1_s` in every operating point table.

Name	Source	Channel	Formula (short)	Original
`van_baten_voll`	van Baten, Krishna 2004	both	cap + film, penetration theory	circular, d 1.5–3 mm, L_UC 15–50 mm
`van_baten_kappe`, `van_baten_film`	ditto	both	contribution of cap or film	
`vandu`	Vandu et al. 2005	both	4.5 √(D u_B / L_UC) / d	d 1–3 mm
`shao`	Shao et al. 2010	both	3.0 √(D u_B / L_UC) / d	CFD, CO2/NaOH
`liu_wang`	Liu, Wang 2011	circular	0.12 Li^0.44 Pe^0.54 D/d²	d 1–3 mm, L_S 8–28 mm
`heiszwolf`	Heiszwolf et al. 2001	both	0.133 u^1.2 / √L_S · (D/D_ref)^0.5	monolith, square
`yao`	Yao et al. 2023	both	van Baten with Ca exponents	0.3 × 0.6 mm
`yue_2007`	Yue et al. 2007	square	0.084 Re_G^0.213 Re_L^0.937 Sc^0.5 D/d²	0.5 × 1 mm, CO2/water
`yue_2009`	Yue et al. 2009	square	(2/d) √(D u_B/d) (L_B/L_UC)^0.3	d 0.2–0.4 mm

Inputs:

- averaged fitted bubble velocity u_B, bubble and slug length as well as d and D(T) of the operating point
- Yue 2007 uses the superficial velocities
- property data for Ca, Re and Sc are fixed (water/CO2, approx. 25 °C)

The supplied channels lie outside the original range of almost all models (larger channels, longer unit cells); the deviations are correspondingly large. Only Yue 2007 follows the trend in the glass channel (slope in the log-log plot 0.92, $R^2 = 0.92$), with about 70 % systematic overestimation.

9.3 Fitting an own model

The block `eigener_fit` in `config.VERGLEICH` fits a power law:

$$y = a \cdot x_1^{b_1} \cdot x_2^{b_2} \cdots$$

- y is the target quantity.
- x_k are any numeric columns of the operating point table, e.g. `GasVelocity_m_s`, `LiquidVelocity_m_s`, `BubbleLength_mean_m`, `SlugLength_mean_m`, `Velocity_mean_m_s`, `d_h_m`.
- The fit is a least-squares fit on log y. This minimises relative deviations.
- The output is the formula, n, MAPE and R^2 in logarithmic space.

```
VERGLEICH = dict(
    ...
    eigener_fit=dict(
        name="pvc_kla",                    # name for later use
        titel="PVC: kLa = a uG^b uL^c",
        kanaele=["PVC_Rund_1000um_72cm_20molCO2_600um",
                 "PVC_Rund_1000um_72cm_20molCO2_800um"],
        variablen=["GasVelocity_m_s", "LiquidVelocity_m_s"],
        speichern=True,                    # save
    ),
)
```

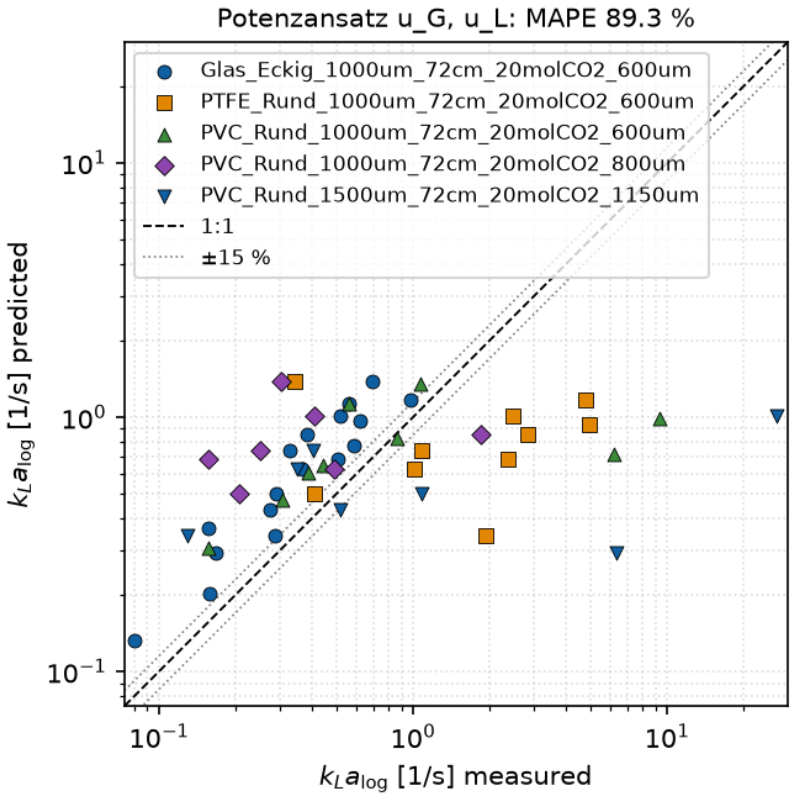


Figure 7: Power law in u_G and u_L, fitted to all channels

With `speichern=True` the model is saved in `config/eigene_modelle.yaml`; an entry of the same name is overwritten. It is then available everywhere by its name, like a literature model:

- in `VERGLEICH["modelle"]`
- in the plot `modell_flaechen` (chapter 4.7)
- in `MONOLITH["vergleich_modell"]` (chapter 11)

Structure of a saved entry (numbers for illustration only):

```
pvc_kla:
  titel: 'PVC: kLa = a uG^b uL^c'
  zielgroesse: kLa_log_mean_1_s
  variablen: [GasVelocity_m_s, LiquidVelocity_m_s]
  koeffizient: 12.3
  exponenten: [0.61, 0.95]
  form: potenz
  datengrundlage: [PVC_Rund_1000um_72cm_20molCO2_600um, PVC_Rund_1000um_72cm_
↪  20molCO2_800um]
```

```
  n_punkte: 16
  mape_prozent: 24.1
  r2_log: 0.81
  erstellt: '2026-10-14'
```

The file may also be edited by hand, e.g. to enter a literature correlation in power-law form with own coefficients.

Other target quantities: any numeric column is possible, e.g. `n_spez_mol_s_m3` or `slip_meas_over_theo`. The literature models all refer to kLa; for other quantities only own models make sense.

9.4 From Python

```
import config
from taylorflow import plots
from taylorflow.modellvergleich import (lade_betriebspunkte_mehrerer_kanaele, fit_
 ↪  potenzansatz,
                                        vergleichstabelle, vorhersage)

daten = lade_betriebspunkte_mehrerer_kanaele(config.ERGEBNIS_ORDNER)
glas = daten[daten["Kanal"].str.startswith("Glas")]

# Another approach: kLa over bubble velocity and unit cell length
glas = glas.assign(L_UC_m=glas["BubbleLength_mean_m"] + glas["SlugLength_mean_m"])
m = fit_potenzansatz(glas, "kLa_log_mean_1_s", ["Velocity_mean_m_s", "L_UC_m"],
 ↪  name="glas_ub_luc")
print(m.formel(), m.mape_prozent)

print(vergleichstabelle(glas, "kLa_log_mean_1_s", ["yue_2007", "vandu"]))

fig = plots.paritaet(glas["kLa_log_mean_1_s"], {"Yue 2007": vorhersage(glas, "yue_
 ↪  2007")},
                     plots.PlotStil(), r"$k_La$ [1/s]", band_prozent=30,
 ↪  logarithmisch=True)
plots.speichere(fig, "abbildungen", "paritaet_yue", plots.PlotStil())
```

A model that uses derived columns such as `L_UC_m` here needs this column again when it is called later. For permanent use via `config.py`, the existing columns of the operating point table are therefore preferable.

9.5 Adding a new literature model as a formula

For models that are not a power law, a function is written in `taylorflow/literatur.py` and entered in `LITERATURMODELLE` (chapter 12.3). The model then appears automatically as a column in the operating point tables (after a new evaluation), in the comparison and as a surface.

10 Pressure calibration

> **Status:** The procedure is fully included but not yet finally validated and is currently not used routinely. The evaluation only reads the existing database `data/calibrated_c_database.json`. `kalibrierung.py` only changes this file if explicitly set to do so.

10.1 Purpose

In the circular channel, the pressure drop is calculated with the Bretherton model (chapter 7.4). It contains a factor c with the theoretical value 0.465. With this value the calculated pressure profile does not fit all channels, and no other published model fits better (chapter 8.2). c is therefore determined per channel from measurements without CO2; there the bubble length changes only because of the pressure.

Square channels (Kurimoto model) are not calibrated.

10.2 Procedure

For each operating point of a file `Auswertung_Druck_…_0molCO2_….csv`:

1. **Fits:** bubble length (weighted), slug length and velocity linearly over z.
2. **Real pressure drop** from the bubble expansion (ideal gas, volume proportional to length):

$$p_{tr,in} = p_{tr,out}\,\frac{L_B(z_{end})}{L_B(0)}, \qquad \Delta p_{real} = p_{tr,in} + p_v - p_{out}$$

3. **Splitting the model:** the Bretherton model is linear in c:
   - $\mathrm{d}p/\mathrm{d}z = \text{base} + c \cdot \text{extra}$
   - base = model with c = 0
   - extra = model(c = 1) − model(c = 0)
   - Both parts are integrated over the channel.
4. **Result:** $c = (\Delta p_{real} - \Delta p_{base})/\Delta p_{extra}$

Over all operating points, three **fallback models** for c(u_G, u_L) are fitted:

Model	Form
Linear	$a\,u_G + b\,u_L + c_0$
Polynomial	$a\,u_G + b\,u_L + c_1\,u_G u_L + c_0$
Power-Law	$a\,u_G^{b}\,u_L^{c_1}$

For each model, R^2 and AIC (smaller = better; penalises additional parameters) are reported.

10.3 Use in the evaluation

For a circular channel, the entry with the calibration key is looked up, e.g. `PVC_circular_1000um_600um` (chapter 2.2). Then:

Situation	c used
within the calibrated operating points	linear interpolation
outside	stored fallback model (`fallback_model`)
no entry, `"USE": false` or fewer than 4 points	theoretical value 0.465 (message in the console)

The result is at least 0.001.

Content of the database:

Key	Points	c	Fallback model	used
`PVC_circular_1000um_600um`	5	0.31–1.33	Power-Law (R^2 0.91)	yes
`PVC_circular_1000um_800um`	8	0.05–0.70	Polynomial (R^2 0.58)	yes
`PVC_circular_1500um_1150um`	8	−7.6 to +16	Polynomial	no (`USE: false`)
`Glas_rectangular_1000um_600um`	8	–	–	no; square, Kurimoto
PTFE	–	–	–	no entry, c = 0.465

An entry can be switched off by hand:

```
"PVC_circular_1500um_1150um": {
    "USE": false,
    "NOTE": "c_calib zwischen -7.6 und +16, vermutlich Messfehler",
    ...
}
```

10.4 New calibration

1. Evaluate the pressure measurement with the video analysis; the file `Auswertung_Druck_….csv` is written to `output/Output_PressureCalibration/`. Material, shape, D and needle must match the mass transfer measurement.
2. In `config.py`, section `KALIBRIERUNG`:

Entry	Default	Meaning
`messdateien` (files)	`"dialog"`	`"dialog"` = selection window, `"alle"` = all `Auswertung_Druck_*.csv` in `ROHDATEN_ORDNER`, or a list of paths
`dialog_ordner`	`ROHDATEN_ORDNER`	start folder of the selection window, e.g. `Output_PressureCalibration` of the video analysis

Entry	Default	Meaning
`ersatzmodell` (fallback model)	`"auto"`	`"auto"` (largest R^2) or `"Linear"`, `"Polynomial"`, `"Power-Law"`
`datenbank_schreiben` (write database)	`False`	leave at `False` first

3. Run `python kalibrierung.py` and select the file(s).
4. Check the results in `ergebnisse/kalibrierung/`:
   - `<key>_c_werte.csv`: c per operating point
   - `<key>_c_flaeche.png`: c over u_G and u_L

   Plausible values are of the order of the theoretical value (0.1–2). Negative or strongly scattering values indicate a problem in the pressure measurement or the fits.
5. Only then set `datenbank_schreiben = True` and run again. Existing fields `USE` and `NOTE` are kept. Then run `auswertung.py` again for the channel.

A new calibration of the supplied pressure data reproduces the stored c values to about 10^{-4} and selects the same fallback models.

10.5 Open questions

- **Assumption volume proportional to length:** caps are neglected, the film thickness is taken as constant. For short bubbles the assumption is rough.
- **Fitting with few points:** five to eight points per channel are few for fitting three to four parameters. `auto` selects by R^2 and thus favours the polynomial; by AIC the linear model would be chosen for PVC 1000/800. For operating points within the calibrated points this does not matter (interpolation).
- **PVC 1500/1150:** the strongly scattering c values are not explained.
- **Measurement deviation of the bubble length:** it enters the calibration only via the standard deviation, not via the standard error and systematic part as in the bubble length fit of the mass transfer measurement.

11 Monolith

`python monolith.py` evaluates measurements on a monolith via a gas balance and compares them with a single channel.

11.1 Measurement data and settings

The measurement file (default `data/raw/Monolith_Messungen.csv`) contains one row per operating point, separated by semicolons, with a decimal point:

Column	Unit	Meaning
`V_L_L_h`	L/h	liquid volume flow
`V_G_in_L_h`, `V_G_out_L_h`	L/h	gas volume flow at inlet and outlet
`y_CO2_in`, `y_CO2_out`	%	CO2 fraction at inlet and outlet (analyser)
`T_in_C`, `T_out_C`	°C	temperature
`p_in_Pa`, `p_out_Pa`	Pa	absolute pressure

Settings in `config.py`, section `MONOLITH`:

Entry	Default	Meaning
`datei` (file)	`data/raw/Monolith_Messungen.csv`	measurement file
`n_kanaele`	100	number of channels
`d_kanal_m`	1e-3	side length of the square channels [m]
`laenge_m`	1.0	length [m]
`henry_ref_mol_m3_pa`	3.3e-4	Henry constant at 298.15 K [mol/(m³·Pa)]
`henry_dlnh_dt_k`	2400	temperature dependence d(ln H)/d(1/T) [K]
`vergleich_kanal` (comparison channel)	glass channel	channel name for the comparison; `None` = no comparison
`vergleich_modell` (comparison model)	`"yue_2007"`	additional model surface; `None` = none

11.2 Calculation

All channels are considered together. Total cross-section $A = N d^2$, volume $V = A\,L$.

Superficial velocities:

$$u_G = \dot{V}_{G,in}/A, \qquad u_L = \dot{V}_L/A$$

Absorbed molar flux from the gas balance, with $c_G = p\,y/(RT)$:

$$\dot{n} = \dot{V}_{G,in}\,c_{G,in} - \dot{V}_{G,out}\,c_{G,out}, \qquad \dot{n}_{spec} = \dot{n}/V$$

Equilibrium, residence time and outlet concentration:

$$C^* = H(T)\,p\,y, \qquad t = L/u_L, \qquad C_{L,out} = \dot{n}\,t/(\dot{V}_L t)$$

kLa with logarithmic concentration difference ($\Delta C_{in} = C^*_{in}$, $\Delta C_{out} = C^*_{out} - C_{L,out}$):

$$k_L a = \frac{\dot{n}}{V_L\,\Delta C_{log}}, \qquad \Delta C_{log} = \frac{\Delta C_{in} - \Delta C_{out}}{\ln(\Delta C_{in}/\Delta C_{out})}$$

The Henry constant follows van't Hoff with the values from `MONOLITH`; they differ slightly from those of the channel evaluation ($3.36 \cdot 10^{-4}$ mol/(m³·Pa), chapter 7.5).

Saturated operating points: if ΔC_out ≤ 0, the liquid is saturated at the outlet and kLa cannot be determined (empty). In the supplied data this applies to 14 of 25 operating points.

11.3 Outputs

In `ergebnisse/monolith/`:

File	Content
`monolith_kennwerte.csv`	u_G, u_L, kLa, ṅ_spec, saturation flag per operating point
`kla_monolith.png`, `nspez_monolith.png`	3D surfaces of the monolith
`kla_kanal.png`, `nspez_kanal.png`	the same quantities for `vergleich_kanal`
`kla_modell.png`	model surface (`vergleich_modell`)
`paritaet_monolith_kanal.png`	kLa of the channel against kLa of the monolith

All 3D surfaces of a run have the same axis ranges and can be compared directly. Language, format and sizes follow `STIL` (chapter 4.10).

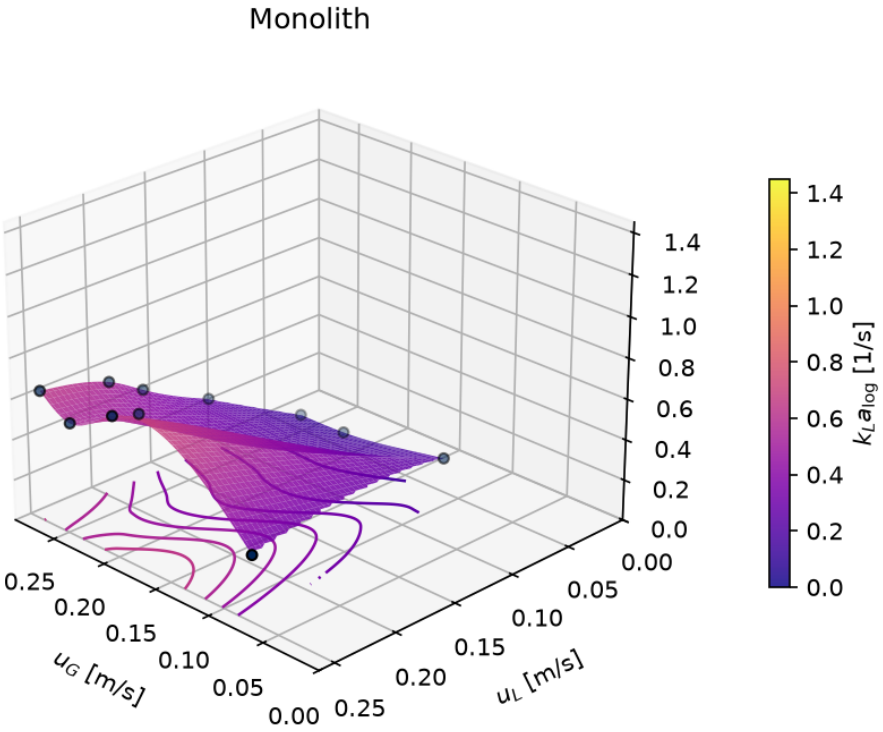


Figure 8: kLa of the monolith

Parity plot:

- The channel values are interpolated to the operating points of the monolith.
- For saturated monolith points, the monolith value is interpolated from its own surface and marked separately.

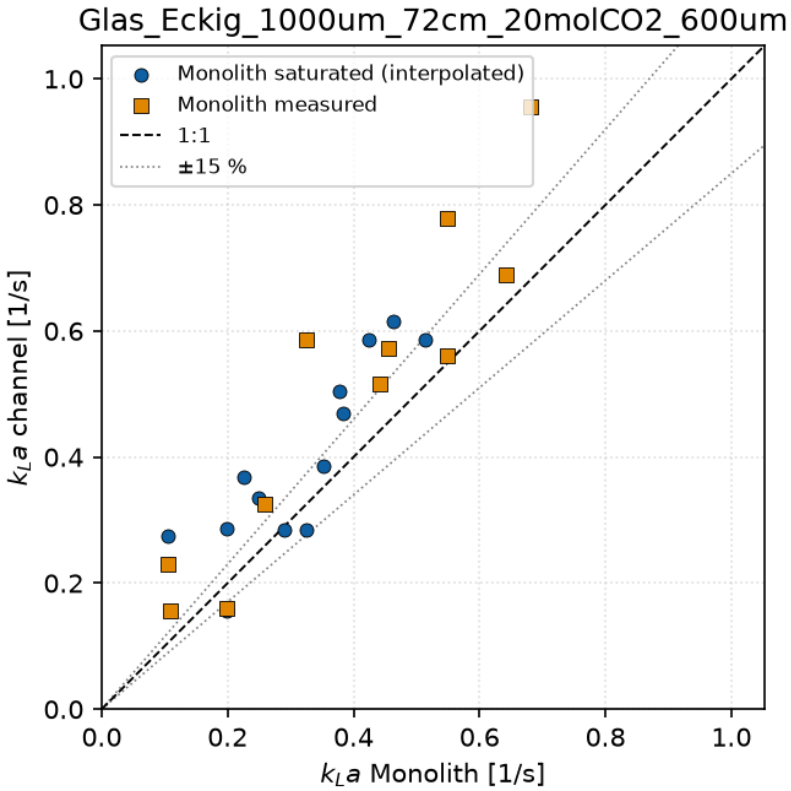


Figure 9: Parity monolith against glass channel

11.4 Interpretation

ṅ_spec of the channel is deliberately related to the whole channel volume, as for the monolith (chapter 7.7). kLa of the channel, on the other hand, is the log-mean over the active zone (chapter 7.6), kLa of the monolith a log-mean over the whole reactor. With early saturation, the two kLa values are therefore only comparable to a limited extent.

12 Program structure and extension

12.1 Structure

```
config.py              all settings
auswertung.py          evaluation and plots
abbildung.py           single figure
literaturvergleich.py  model comparison
kalibrierung.py        pressure calibration
monolith.py            monolith
taylorflow/
  parameter.py         numbers of the evaluation (dataclass AuswerteParameter)
  einlesen.py          file selection, file names, data, exclusions, camera filter
  pipeline.py          evaluation per channel and operating point (steps 1–8,
   ↪  chapter 6.1)
  fits.py              bubble model with camera offsets, linear fits
  druckverlust.py      Bretherton, Kurimoto, calibration database
  hydrodynamik.py      capillary number, film thickness
  geometrie.py         bubble volume and cross-section
  stoffdaten.py        thermo interface, Henry, vapour pressure, diffusion
  kla.py               active zone, kLa_log_mean
  unsicherheit.py      error propagation (not called by the evaluation)
  literatur.py         literature models and registry
  modellvergleich.py   predictions, power-law fit, statistics, own models
  ergebnisse.py        saving and loading the result files
  protokoll.py         warning log
  monolith.py          gas balance of the monolith
  kalibrierung.py      pressure calibration
  plots/
    stil.py            PlotStil, German/English texts, saving
    optionen.py        options per plot: defaults, checks, calls
    profile.py         plots per operating point
    flaechen.py        3D surfaces
    heatmaps.py        key figure overview
    paritaet.py        parity plots
tests/                 tests and reference results
docs/                  manual (Markdown), figures, PDF generation
```

Dependencies: the scripts in the project folder read `config.py` and call the package `taylorflow`. The package itself does not know `config.py`; all settings are passed as arguments. The functions can therefore also be used individually.

12.2 Tests

```
python -m pytest
```

File	Checks
`tests/test_regression.py`	evaluates all supplied measurement files and compares every number of the operating point tables and an extract of the profiles exactly with `tests/referenz/`
`tests/test_bausteine.py`	single parts with synthetic data: file names, fit, active zone, kLa_log_mean, uncertainty, model registry, power law
`tests/test_dateiauswahl.py`	selection of measurement files (list, “alle”, selection window), files of the video analysis
`tests/test_plots.py`	plot options in `config.py` and `abbildung.py` for every plot

After every code change, the regression test shows whether results have changed.

If a change alters the results on purpose (new model, correction):

1. Quantify the effect:

   ```
   python tests/referenz_erneuern.py
   ```

   lists every differing column with the largest relative deviation.

2. Rewrite the reference:

   ```
   python tests/referenz_erneuern.py --schreiben
   ```

3. Increase `__version__` in `taylorflow/__init__.py`; the version is stored in every `<channel>_meta.json`.

12.3 Typical extensions

12.3.1 New literature model

In `taylorflow/literatur.py`:

```python
def mein_modell(u_B, L_B, L_S, d_h, D):
    """Author (year), journal. Valid for …"""
    return 2.0 * np.sqrt(D * u_B / (L_B + L_S)) / d_h

LITERATURMODELLE["mein_modell"] = Literaturmodell(
    "mein_modell", "Author 2027", "kLa_lit_mein_modell_1_s",
    lambda x: mein_modell(*_e(x, "u_B", "L_B", "L_S", "d_h", "D")))
```

The available inputs are listed in `modelleingaben()`. New quantities are added there and in the calls in `pipeline.py` and `modellvergleich.vorhersage`. The new column changes the operating point table; rewrite the reference afterwards (12.2).

12.3.2 New plot

1. Create a function in a module under `taylorflow/plots/`. It receives the data, a `PlotStil`, a title and its options as keyword arguments and returns a figure. Building blocks:
   - `neue_figur(stil, groesse_cm)`
   - `t(schluessel, stil)` for labels; enter new texts in `TEXTE`, in German and English
   - `formatiere_achsen(ax, stil)`
2. In `taylorflow/plots/optionen.py`:
   - an entry in `STANDARD` with all options and defaults
   - a branch in `erzeuge` that calls the function and sets the file name
3. Create a block with the same entries as in `STANDARD` in `config.PLOTS`. The test `test_config_zeigt_alle_optionen_mit_standardwerten` checks that both match.

The plot is then available in `auswertung.py` and `abbildung.py`.

12.3.3 New option for an existing plot

Add a parameter with a default value to the plot function, and enter the same name with the same default in `STANDARD` (`optionen.py`) and in the block in `config.PLOTS`.

12.3.4 Other liquid or gas

- **Liquid:** set `fluessigkeit` in `config.PARAMETER` (name for thermo, e.g. `"ethanol"`).
- **Henry constant:** for liquids other than water, extend `stoffdaten.henry`; currently an error message is raised.
- **Diffusion coefficient:** `stoffdaten.diffusion_co2_wasser` only applies to CO2/water. It is used only for the literature models and the column `D_CO2_m2_s`.
- **Gas:** enter new gases in `HENRY_DATEN`.

12.3.5 New evaluation parameter

1. Add a field with a default value in `AuswerteParameter` (`parameter.py`).
2. Use it in the calculation via `p.<name>`.
3. List it in `config.PARAMETER` and document it in chapter 5.

12.4 Building the manual

The manual consists of the Markdown files in `docs/` (German) and `docs/en/` (English). Figures and PDF are created with two scripts:

```
python docs/bilder_erzeugen.py          figures from the supplied data (German
↪  labels)
python docs/bilder_erzeugen.py en       the same with English labels
python docs/build_pdf.py                docs/Handbuch_TaylorFlow.pdf
python docs/build_pdf.py en             docs/Handbuch_TaylorFlow_en.pdf
```

The PDF additionally requires `pip install pypandoc_binary` and a LaTeX installation with XeLaTeX (TeX Live or MiKTeX).